

ARITHMETIC FORMS AND OPERATOR DOMAINS

Closing an operator kernel to a scalar and giving states positive norms are separate requirements in a field theory. A cyclic trace controls an operator contraction. An involution and a positive functional instead produce a Hilbert space and operators on a common domain. Arithmetic supplies a concrete test of the second construction: can its prime, archimedean and polar terms define a positive pairing without prescribing the spectrum in advance?

The arithmetic terms of the zeta explicit formula define a Hermitian pairing on a nonunital convolution algebra. Gaussian reflection and a contour integral derive the full formula on this algebra. A separate polynomial–Gaussian argument identifies its positivity with the Riemann hypothesis. A local Gaussian calculation proves a further restriction: the full Weil functional admits no positive unitization of finite mass and no realization by a finite-norm vector on an invariant $*$ -representation domain. The same calculation gives positive Gaussian subspaces of every finite dimension at sufficiently small scales. We construct the common graph completion of nonunital GNS multipliers and prove its exact domain and transfer properties. Closed quadratic forms for individual multipliers need not have invariant domains. Their common domain over the whole algebra does. An independent positive arithmetic functional remains open, with its possible representation constrained by the Gaussian divergence.

The decisive question is whether the arithmetic expression is a squared norm for a reason independent of the location of the zeros. Such a norm would construct a Hilbert space from test functions. It need not come from a distinguished vector representing the functional. The Gaussian calculation in Section 1.16 excludes that stronger realization. We first derive the arithmetic expression and its positivity criterion; then we construct the representation and its domain under a stated positivity hypothesis.

1.1 THE ARITHMETIC IDENTITY AND ITS VARIABLES

The Dirichlet series $\sum_{n \geq 1} n^{-s}$ has two useful descriptions. Prime factorization gives a product when $\operatorname{Re} s > 1$. A Gaussian sum gives a continuation and a reflection in $s = 1/2$. Taking a logarithmic derivative changes prime factors into weighted prime powers and changes zeros into simple poles. A contour integral can then compare them. The test function must control that contour before it can define an identity.

All vector spaces are complex, and all real integrals use Lebesgue measure. The starting language is calculus, absolutely convergent integrals and series, and the complete fields $\mathbb{R}$ and $\mathbb{C}$. We use the usual limit rules for absolutely integrable functions: an integrable common bound permits passage through an integral. Every infinite sum and parameter integral below has an explicit such bound. The elementary complex-variable tools needed for the contour are recorded in Section 1.2.

For $a > 0$, define

$$\begin{aligned} \mathcal{A}_a &= \left\{ g \in C^\infty(\mathbb{R}) : p_{a,k}(g) := \int_{\mathbb{R}} e^{a|u|} |g^{(k)}(u)| du < \infty \text{ for every } k \geq 0 \right\}, \\ \mathcal{A} &= \bigcup_{a > 1/2} \mathcal{A}_a. \end{aligned} \tag{1.1}$$

Here C^∞ means that every real derivative exists and is continuous. The functions may be complex-valued and need not be even. Define convolution, involution, and Fourier transform by

$$(f \star g)(u) = \int_{\mathbb{R}} f(v) g(u - v) dv, \quad g^*(u) = \overline{g(-u)}, \quad \widehat{g}(z) = \int_{\mathbb{R}} g(u) e^{izu} du. \quad (1.2)$$

The complex variable suited to the Dirichlet series is

$$F_g(s) = \widehat{g}(-i(s - 1/2)) = \int_{\mathbb{R}} g(u) e^{(s-1/2)u} du = \int_0^\infty g(\log x) x^{s-1/2} \frac{dx}{x}. \quad (1.3)$$

The last equality is the change of variable $x = e^u$. In particular $F_g(1/2 + it) = \widehat{g}(t)$. The half in this substitution determines the weights of all prime terms.

Write $\Lambda_{\text{vM}}(n) = \log p$ if $n = p^m$ with p prime and $m \geq 1$, and write $\Lambda_{\text{vM}}(n) = 0$ otherwise. The gamma function and the meromorphic continuation of ζ will be constructed below. Put $\psi = \Gamma'/\Gamma$. The identity to be proved is

$$\begin{aligned} \sum_{\rho} F_g(\rho) &= F_g(0) + F_g(1) \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) \left[\operatorname{Re} \psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi \right] dt \\ &- \sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (g(\log n) + g(-\log n)). \end{aligned} \quad (1.4)$$

Zeros ρ are counted with multiplicity. Their precise definition will exclude the canceled gamma poles and the poles at zero and one. Theorem 1.7.1 proves (1.4) for every $g \in \mathcal{A}$. The proof uses no positivity statement.

1.2 ELEMENTARY COMPLEX ESTIMATES

A holomorphic function is a complex-differentiable function on an open set. Differentiability implies continuity, but we will not assume continuity of the derivative in deriving the integral formulas. A meromorphic function is locally a quotient of holomorphic functions, with denominator not identically zero. A zero or pole has order m if its local expression contains a factor $(z - z_0)^m$ or $(z - z_0)^{-m}$ with nonzero remaining factor. The residue is the coefficient of $(z - z_0)^{-1}$.

For a piecewise continuously differentiable path $\gamma : [a, b] \rightarrow \mathbb{C}$, define

$$\int_{\gamma} h(w) dw = \int_a^b h(\gamma(t)) \gamma'(t) dt,$$

adding the integrals over its differentiable pieces. Reversing a path reverses the sign of its integral.

LEMMA 1.2.1 (*Integration around a triangle*). If h is holomorphic on a neighborhood of a closed triangle T , then $\int_{\partial T} h(w) dw = 0$, with the boundary oriented counterclockwise.

Proof. Joining the edge midpoints divides T into four triangles. Their oriented boundary integrals sum to the original integral because every internal edge occurs in both directions. Choose a subtriangle whose integral has modulus at least one quarter of that of its parent, and repeat. The resulting nested closed triangles T_n have diameter $2^{-n}D$ and perimeter $2^{-n}P$, where D, P are those of T . Points chosen in

these triangles form a Cauchy sequence. Completeness gives a limit w_o belonging to every T_n , and their shrinking diameters make this common point unique.

Differentiability at w_o gives

$$h(w) = h(w_o) + h'(w_o)(w - w_o) + (w - w_o)\epsilon(w), \quad \epsilon(w) \rightarrow 0 \quad (w \rightarrow w_o).$$

The first two terms have polynomial primitives, so their integrals around each triangle vanish by the real fundamental theorem of calculus on each edge. Put $\epsilon_n = \sup_{w \in T_n} |\epsilon(w)|$, defining $\epsilon(w_o) = 0$. Then $\epsilon_n \rightarrow 0$, and

$$\left| \int_{\partial T} h(w) dw \right| \leq 4^n \left| \int_{\partial T_n} h(w) dw \right| \leq 4^n \epsilon_n (2^{-n} D) (2^{-n} P) = \epsilon_n D P \rightarrow 0.$$

□

LEMMA I.2.2 (*Contour formulas*). Let a function be holomorphic on a neighborhood of a closed disc. For any interior point z ,

$$h(z) = \frac{1}{2\pi i} \int_{|w-z_o|=R} \frac{h(w)}{w-z} dw, \quad |h^{(k)}(z_o)| \leq \frac{k!}{R^k} \max_{|w-z_o|=R} |h(w)|.$$

The contour has counterclockwise orientation. For a meromorphic function on a rectangle with no boundary poles, its boundary integral is $2\pi i$ times the sum of its interior residues. Locally uniform limits of holomorphic functions are holomorphic.

Proof. Summing the triangle identity gives zero around any polygonal region on whose neighborhood the integrand is holomorphic, including a region with polygonal holes. To obtain a finite subdivision, draw parallel lines through its finitely many vertices, divide the resulting strips into trapezoids and triangles, and split each trapezoid along a diagonal. The internal edges cancel, leaving the outer boundary counterclockwise and the inner boundaries clockwise.

The same identity holds for a circular boundary by polygonal approximation. Linear interpolation of its smooth parametrization converges uniformly in position; on each interpolation interval its derivative is the average of the original derivative. Uniform continuity of that derivative gives convergence of tangents in the integral norm. Continuity of the integrand, uniformly on a compact neighborhood of the circle, then gives convergence of the boundary integrals.

Apply this identity to $h(w)/(w-z)$ between the original circle and a small circle centered at z . Its neighborhood contains no singularity of this integrand. On the small circle, write $h(w) = h(z) + O(|w-z|)$ and let its radius tend to zero. The constant term integrates to $2\pi i h(z)$ and the error tends to zero. This proves the first formula without a regularity assumption on h' . Expansion of $(w-z)^{-1}$ as a geometric series gives the derivative formula and the displayed estimate, as well as a convergent local power series for every holomorphic function. For the residue formula, remove disjoint small discs around the poles. The same boundary argument applies to the remaining region. The finite principal part at each pole follows by factoring the first nonzero Taylor term of the denominator in a local quotient. On its small circle, only the term $(w-w_o)^{-1}$ has nonzero integral, namely $2\pi i$ times its coefficient. Adding the circles proves the residue statement. For a locally uniform limit, pass to the limit in the circle formula. The resulting expression is differentiable in the interior point and gives its convergent power series. These arguments also justify the formulas on discs and rectangles with finitely many small discs removed. □

The circle formula implies the identity theorem: a nonzero holomorphic function has isolated zeros, since its first nonzero Taylor coefficient factors out the entire order of a zero. It also implies the maximum modulus principle. If an interior modulus is maximal, equality in the circle average forces constant argument and modulus on a small circle. Its Taylor coefficients of positive degree then vanish, and the function is constant on the connected domain.

LEMMA 1.2.3 (*Logarithms and counting zeros*). A zero-free holomorphic function on a disc has a holomorphic logarithm. If an entire function h satisfies $h(o) \neq 0$ and

$$\log \max_{|z| \leq R} |h(z)| \leq CR \log(R+2) \quad (R \geq 2),$$

then the number $n_h(R)$ of its zeros in $|z| \leq R$, with multiplicity, satisfies $n_h(R) = O(R \log(R+2))$.

Proof. For a zero-free function, the integral of h'/h around any closed contour in the disc is zero by Lemma 1.2.2. Its path integral is thus a primitive ℓ . Differentiation shows that $e^{-\ell}h$ is constant. Choosing the additive constant gives $h = e^\ell$.

Choose $r > R$ so that h has no zero on $|z| = r$. Factor out its finitely many zeros a_j inside that circle. The remaining nonzero factor has a logarithm whose real part has its value at zero equal to its circle average. For $|a| < r$, the power series of $\log(1 - ae^{-i\theta}/r)$ gives the circle average $\log |re^{i\theta} - a| = \log r$. Subtracting the value at zero gives

$$\sum_{|a_j| < r} \log \frac{r}{|a_j|} = \frac{1}{2\pi} \int_0^{2\pi} \log |h(re^{i\theta})| d\theta - \log |h(o)|.$$

Take $2R < r < 2R+1$, avoiding the isolated radii of zeros. Each zero in $|z| \leq R$ contributes at least $\log 2$. The asserted growth bound proves the count. $\square$

The contour will need a logarithmic derivative estimate, not only a count. The next lemma supplies that extra information locally.

LEMMA 1.2.4 (*A local logarithmic derivative bound*). Let h be holomorphic on a neighborhood of $|z| \leq R$, where $5 \leq R \leq 6$, and suppose $h(o) \neq 0$ and h has no boundary zero. Suppose

$$|h(z)| \leq e^M \quad (|z| \leq R), \quad |h(o)| \geq e^{-M}, \quad M \geq 1.$$

If h has N zeros in the disc and $|z| \leq 3$ is at distance at least $\delta > 0$ from each zero, then

$$\left| \frac{h'(z)}{h(z)} \right| \leq CM + N(\delta^{-1} + 1/2),$$

where C is an absolute constant.

Proof. For each zero a , including multiplicities, use the factor $b_a(z) = R(z-a)/(R^2 - \bar{a}z)$. It has modulus one on $|z| = R$ and no denominator zero in the disc. Divide h by their product and remove the canceled singularities. The quotient v is holomorphic and zero-free. The maximum principle gives $|v| \leq e^M$. Since $|b_a(o)| < 1$, also $|v(o)| \geq |h(o)| \geq e^{-M}$. Choose $\ell = \log v$ with $\ell(o)$ real after multiplying v by a constant of modulus one. The function $p = M+1-\ell$ has positive real part and $1 \leq p(o) \leq 2M+1$.

The function $q = (p - p(o))/(p + p(o))$ maps the disc into the unit disc and vanishes at zero. Applying the maximum principle to $q(z)/z$ on smaller circles and increasing their radii to R gives $|q(z)| \leq |z|/R$. Hence $|p(z)| \leq 9p(o)$ on $|z| \leq 4$. The derivative estimate on radius-one circles about $|z| \leq 3$ yields $|\ell'(z)| = |p'(z)| \leq 9p(o) \leq 27M$. Finally

$$\frac{h'}{h} = \ell' + \sum_a \left(\frac{1}{z-a} + \frac{\bar{a}}{R^2 - \bar{a}z} \right).$$

For $|z| \leq 3$, the second fraction has modulus at most $1/(R-3) \leq 1/2$. The assumed distance bound controls the first. $\square$

1.3 THE TEST ALGEBRA AND GAUSSIAN FOURIER INVERSION

LEMMA 1.3.1 (*Decay, boundary values, and transforms*). If $g \in \mathcal{A}_a$, then $e^{a|u|} g^{(k)}(u) \rightarrow 0$ at both ends of $\mathbb{R}$, and $|g^{(k)}(u)| \leq C_k e^{-a|u|}$. The transform $\widehat{g}$ is holomorphic for $|\operatorname{Im} z| < a$. For every $0 < b < a$ and every integer $N \geq 0$,

$$\sup_{|\operatorname{Im} z| \leq b} (1 + |z|)^N |\widehat{g}(z)| < \infty. \quad (1.5)$$

Equivalently, F_g is holomorphic for $1/2 - a < \operatorname{Re} s < 1/2 + a$ and decreases faster than any inverse power of $|\operatorname{Im} s|$ on every smaller closed vertical strip.

Proof. On $u \geq 0$, the function $h(u) = e^{au} g^{(k)}(u)$ and its derivative $e^{au}(a g^{(k)} + g^{(k+1)})$ are integrable. The integral of the derivative makes h have a finite limit at infinity. Integrability of h forces that limit to be zero. Moreover $|h(u)| \leq \int_u^\infty |h'(v)| dv$. The negative half-line has the same proof after reflection.

For $b < a$, the functions $|u|^m e^{b|u|} |g(u)|$ are integrable for every m , because $|u|^m e^{-(a-b)|u|}$ is bounded. These bounds permit termwise complex differentiation of the integral. Repeated integration by parts has zero boundary terms by the first assertion and gives

$$(-iz)^N \widehat{g}(z) = \int_{\mathbb{R}} g^{(N)}(u) e^{izu} du.$$

Its right side is bounded on $|\operatorname{Im} z| \leq b$. Combining this with the bound for $N = 0$ proves (1.5). Substitution in (1.3) proves the last assertion. $\square$

LEMMA 1.3.2 (*The normalized Gaussian*). For $A > 0$, put

$$g_A(u) = (4\pi A)^{-1/2} e^{-u^2/(4A)}.$$

Then $g_A \in \mathcal{A}_a$ for every $a > 0$, and

$$\widehat{g}_A(z) = e^{-Az^2}, \quad g_A \star g_B = g_{A+B}, \quad \int_{\mathbb{R}} g_A(u) du = 1. \quad (1.6)$$

For $g \in \mathcal{A}$ one has the inverse formula

$$g(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) e^{-itu} dt. \quad (1.7)$$

The same inverse formula holds for a smooth function g with $g, g', g'' \in L^1(\mathbb{R})$.

Proof. Squaring $\int e^{-x^2} dx$ and using polar coordinates gives its value $\sqrt{\pi}$. Scaling proves the mass assertion. Every derivative of g_A is a polynomial times this Gaussian, proving membership and vanishing of every exponential-weighted boundary value. For real t , differentiate $J(t) = \widehat{g}_A(t)$ under the integral. Since $u g_A = -2A g'_A$, integration by parts gives $J'(t) = -2At J(t)$. The initial value $J(0) = 1$ gives $J(t) = e^{-At^2}$. Both sides are entire, so the identity theorem extends the formula to $z \in \mathbb{C}$. Completing the square in the convolution integral gives g_{A+B} .

For inversion, insert the factor e^{-At^2} on the right of (1.7). Absolute integrability permits reversing the integrals. The Gaussian transform just proved gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-At^2} \widehat{g}(t) e^{-itu} dt = (g_A \star g)(u).$$

The left side tends to the proposed inverse by (1.5). On the right, continuity of g controls $|v| < \eta$ in the convolution difference, and the Gaussian mass outside $|v| < \eta$ tends to zero. Boundedness of g controls that tail. Thus $(g_A \star g)(u) \rightarrow g(u)$. For the stated larger class, integrability of g, g', g'' makes g, g' bounded with zero limits at infinity, by the half-line argument of Lemma 1.3.1 with zero weight. Two integrations by parts give an integrable bound for $\widehat{g}$. The same proof then applies. $\square$

The parameter A is also a heat parameter. Direct differentiation gives

$$\partial_A g_A(u) = \left(-\frac{1}{2A} + \frac{u^2}{4A^2} \right) g_A(u) = \partial_u^2 g_A(u).$$

Thus convolution by g_A solves heat evolution on the real line for the smooth integrable initial functions considered here. The factor e^{-At^2} damps high Fourier frequencies, and $g_A \star g_B = g_{A+B}$ expresses addition of elapsed heat times. This supplies an explicit analytic regularization for the inverse transform and for the arithmetic identities below.

A complex $\ast$ -algebra is an associative complex algebra with a conjugate-linear map $a \mapsto a^*$ such that $(a^*)^* = a$ and $(ab)^* = b^* a^*$. For the $\ast$ -algebras in this chapter, a unit is a nonzero element 1 with $1a = a1 = a$ for every a ; nonunital means that no such element is present. In particular, the zero algebra is nonunital under this convention.

PROPOSITION 1.3.3 (*Algebra and change of exponential convention*). Each $\mathcal{A}_a$ for $a > 0$, and also $\mathcal{A}$, is a commutative nonunital $\ast$ -algebra under (1.2), and

$$\widehat{f \star g}(z) = \widehat{f}(z) \widehat{g}(z), \quad \widehat{f^*}(z) = \overline{\widehat{f}(\bar{z})}. \quad (1.8)$$

These equations hold on every common strip of holomorphy. The union in (1.1) is unchanged if weighted L^1 bounds are replaced by weighted supremum bounds, with the exponential rate allowed to decrease strictly.

Proof. For $a > 0$ and $f, g \in \mathcal{A}_a$, use $e^{a|u|} \leq e^{a|v|} e^{a|u-v|}$ to obtain $p_{a,k}(f \star g) \leq p_{a,k}(f) p_{a,0}(g)$. If $f, g \in \mathcal{A}$, they admit a common $a > 1/2$. Differentiation, reassociation, and exchange of factors are justified by these absolute integral bounds. Reflection proves the involution rule, and reversing the absolutely convergent double integral proves the first transform equation. The second follows from $u \mapsto -u$ and conjugation. If an identity e existed in any one of these algebras, convolution with g_A would imply $\widehat{e}(t) = 1$, contradicting (1.5).

Lemma 1.3.1 changes weighted L^1 bounds into weighted supremum bounds at the same rate. Conversely, if $|g^{(k)}(u)| \leq C_k e^{-a|u|}$, then for $1/2 < b < a$, $p_{b,k}(g) \leq 2C_k/(a-b)$. This proves the union equality without identifying its individual fixed-rate spaces or silently removing the loss in the rate. $\square$

1.4 PRIME FACTORS AND THE GAMMA FACTOR

For positive x and complex s , write $x^s = \exp(s \log x)$ with the real logarithm of x . This convention fixes every power used below.

LEMMA 1.4.1 (*The convergent prime product*). The series $\zeta(s) = \sum_{n \geq 1} n^{-s}$ defines a holomorphic nonzero function for $\operatorname{Re} s > 1$. In that half-plane,

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \quad \frac{\zeta'(s)}{\zeta(s)} = - \sum_{n \geq 2} \Lambda_{\text{vM}}(n) n^{-s}. \quad (1.9)$$

Both the logarithmic series and its derivative converge absolutely on every smaller closed half-plane. On $s = 2 + it$, $|\zeta(s)| \geq \zeta(2)^{-1}$.

Proof. Existence and uniqueness of prime factorization follow by induction from the fact that a prime dividing a product divides a factor. The latter follows from the integer Euclidean algorithm and its identity for the greatest common divisor. Expanding a finite prime product into geometric series therefore gives the sum over integers supported on those primes. For $\sigma > 1$, the integral comparison $\sum n^{-\sigma} \leq 1 + \int_1^\infty x^{-\sigma} dx$ is finite. It bounds the omitted tails and proves the product identity.

The series $\sum_p \sum_{m \geq 1} p^{-ms}/m$ and its derivative are locally uniformly absolutely convergent: their absolute values are bounded by constants times $\sum_{n \geq 2} (1 + \log n) n^{-\sigma}$. Exponentiating that series gives the product, so the product is nonzero. Differentiation gives (1.9). Finally

$$|\zeta(2 + it)|^{-1} \leq \prod_p (1 + p^{-2}) \leq \prod_p (1 - p^{-2})^{-1} = \zeta(2).$$

□

LEMMA 1.4.2 (*Gamma product and vertical bounds*). For $\operatorname{Re} z > 0$, the integral

$$\Gamma(z) = \int_0^\infty e^{-x} x^{z-1} dx$$

is holomorphic and nonzero. Its reciprocal extends to the entire function

$$E(z) = \frac{1}{\Gamma(z)} = z e^{\gamma z} \prod_{n \geq 1} (1 + z/n) e^{-z/n}, \quad (1.10)$$

where $\gamma = \lim_{N \rightarrow \infty} (\sum_{n=1}^N 1/n - \log N)$. Its zeros are simple and are exactly 0, -1, -2, ... For $R \geq 2$,

$$|E(z)| \leq \exp(CR \log(R+2)) \quad (|z| \leq R). \quad (1.11)$$

The logarithmic derivative satisfies

$$\psi(z) = -\gamma + \sum_{n \geq 0} \left(\frac{1}{n+1} - \frac{1}{n+z} \right), \quad \psi(\sigma + it) = O_{\alpha, \beta}(\log(2 + |t|)) \quad (1.12)$$

uniformly for $0 < \alpha \leq \sigma \leq \beta$.

Proof. The integral and its derivatives have integrable majorants on compact subsets of $\operatorname{Re} z > 0$. Repeated real integration by parts gives

$$\int_0^1 x^{z-1}(1-x)^N dx = \frac{N!}{z(z+1)\cdots(z+N)}.$$

After multiplying by N^z and substituting $u = Nx$, the integrand tends to $u^{z-1}e^{-u}$. The inequality $(1-u/N)^N \leq e^{-u}$ for $0 \leq u < N$ is an integrable bound, also after any fixed number of z derivatives on compact subsets. Thus

$$\Gamma(z) = \lim_{N \rightarrow \infty} \frac{N^z N!}{z(z+1)\cdots(z+N)}.$$

The sequence $\sum_{n=1}^N 1/n - \log N$ decreases and is bounded below by the comparison of $1/x$ with its left and right rectangle sums. Taking reciprocals of the finite expressions gives the finite products in (1.10). For $|w| \leq 1/2$, the power series of the logarithm gives $|\log(1+w) - w| \leq C|w|^2$. Hence their tails converge uniformly on compact sets to a nonzero factor. The finite initial factors identify the zeros and their orders. Multiplication of the two limits gives $\Gamma E = 1$ on the right half-plane, proving nonvanishing and the reciprocal formula.

For $|z| \leq R$, split the product at $n = \lceil 2R \rceil$. The logarithms of absolute values of the initial factors are bounded above by

$$\sum_{n \leq \lceil 2R \rceil} (\log(1 + R/n) + R/n) = O(R \log(R+2)).$$

The remaining logarithmic tail is $O(R^2 \sum_{n > 2R} n^{-2}) = O(R)$. Including $ze^{\gamma z}$ proves (1.11).

Logarithmic differentiation gives the series in (1.12). For $|t| \geq 1$, split that series at $n = \lceil 2(|t| + \beta + 1) \rceil$. The sum of $1/(n+1)$ before that point is $O(\log(2 + |t|))$, and the sum of $|n+z|^{-1}$ is $O(1)$ because $|n+z| \geq |t|$. After that point, each difference is bounded by $C|z-1|/(n+1)^2$, whose sum is $O(1)$. The compact part with $|t| \leq 1$ has no pole and is bounded. $\square$

Define

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2), \quad H_{\Gamma}(s) = \frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2). \quad (1.13)$$

The half in $s/2$ and the real factor $\pi^{-s/2}$ will produce respectively the argument $1/4 + it/2$ and the term $-\log \pi$ in (1.4).

1.5 GAUSSIAN REFLECTION AND THE COMPLETED FUNCTION

The completion of the Dirichlet series will come from a lattice sum. The identity needed for that sum is a direct consequence of the Gaussian Fourier transform, with its normalization kept explicit.

The normalization in Lemma 1.3.2 also gives the Fourier integral used by periodization. The substitution $A = 1/(4\pi x)$ and $z = -2\pi t$ relates the two conventions.

LEMMA 1.5.1 (*Gaussian Fourier transform*). For $x > 0$ and real t ,

$$\int_{\mathbb{R}} e^{-\pi x v^2} e^{-2\pi i t v} dv = x^{-1/2} e^{-\pi t^2/x}.$$

Proof. Squaring $I = \int_{\mathbb{R}} e^{-u^2} du$ and using polar coordinates gives $I^2 = 2\pi \int_0^\infty r e^{-r^2} dr = \pi$. The integrand is positive, so $I = \sqrt{\pi}$. Scaling gives $\int_{\mathbb{R}} e^{-\pi x v^2} dv = x^{-1/2}$. Let $J(t)$ denote the integral in the statement. Every fixed power of $|v|$ times $e^{-\pi x v^2}$ is integrable. This bound permits differentiation in t and integration by parts in v . Using $\partial_v e^{-\pi x v^2} = -2\pi x v e^{-\pi x v^2}$ gives

$$J'(t) = \frac{i}{x} \int_{\mathbb{R}} \partial_v (e^{-\pi x v^2}) e^{-2\pi i t v} dv = -\frac{2\pi t}{x} J(t).$$

The boundary terms vanish because of Gaussian decay. The initial value $J(0) = x^{-1/2}$ solves this differential equation and proves the formula. $\square$

LEMMA 1.5.2 (*Periodization of a Gaussian*). For $x > 0$, put $\theta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}$. Then

$$\theta(x) = x^{-1/2} \theta(1/x). \quad (1.14)$$

For $x \geq 1$, one has $0 < \theta(x) - 1 \leq C e^{-\pi x}$.

Proof. The periodized function $P_x(u) = \sum_{n \in \mathbb{Z}} e^{-\pi x(u+n)^2}$ and all its derivatives converge uniformly for $0 \leq u \leq 1$. Its k th Fourier coefficient, with period one, is

$$\int_0^1 P_x(u) e^{-2\pi i k u} du = \int_{\mathbb{R}} e^{-\pi x v^2} e^{-2\pi i k v} dv = x^{-1/2} e^{-\pi k^2/x},$$

by Lemma 1.3.2. These coefficients are absolutely summable and define a continuous periodic Fourier series.

For completeness, a continuous periodic function with all Fourier coefficients zero is zero. Indeed, the kernel

$$K_N(u) = \frac{1}{N} \left| \sum_{j=0}^{N-1} e^{2\pi i j u} \right|^2$$

is nonnegative, has integral one over a period, and is a finite Fourier sum. Away from the integers it is at most $1/(N \sin^2 \pi u)$. Uniform continuity and this bound show that convolution with K_N converges uniformly to any continuous periodic function. When all coefficients vanish, each convolution is zero. Apply this uniqueness statement to P_x minus its displayed Fourier series, then set $u = 0$ to obtain (1.14). Finally $\theta(x) - 1 = 2e^{-\pi x} \sum_{n \geq 1} e^{-\pi(n^2-1)x}$, and the last sum is bounded by its value at $x = 1$. $\square$

THEOREM 1.5.3 (*Continuation and reflection*). The function $\mathcal{L}(s) = \Gamma_{\mathbb{R}}(s) \zeta(s)$, initially defined for $\operatorname{Re} s > 1$, extends meromorphically to $\mathbb{C}$ by

$$\mathcal{L}(s) = -\frac{1}{s} + \frac{1}{s-1} + \frac{1}{2} \int_1^\infty (\theta(x) - 1) \left(x^{s/2} + x^{(1-s)/2} \right) \frac{dx}{x}. \quad (1.15)$$

It has simple poles at zero and one, with residues -1 and 1 . It satisfies $\mathcal{L}(s) = \mathcal{L}(1-s)$ and $\mathcal{L}(\bar{s}) = \overline{\mathcal{L}(s)}$. The function

$$X(s) = \frac{1}{2} s(s-1) \mathcal{L}(s) \quad (1.16)$$

is entire, obeys the same symmetries, and satisfies $X(0) = X(1) = 1/2$ and

$$\log \max_{|s| \leq R} |X(s)| \leq C R \log(R+2) \quad (R \geq 2). \quad (1.17)$$

Every zero of X lies in $0 \leq \operatorname{Re} s \leq 1$.

Proof. For $\operatorname{Re} s > 1$, absolute convergence gives

$$\frac{1}{2} \int_0^\infty (\theta(x) - 1) x^{s/2} \frac{dx}{x} = \sum_{n \geq 1} \int_0^\infty e^{-\pi n^2 x} x^{s/2} \frac{dx}{x} = \Gamma_{\mathbb{R}}(s) \zeta(s).$$

For the integral from zero to one, use $\theta(x) - 1 = x^{-1/2} - 1 + x^{-1/2}(\theta(1/x) - 1)$. The first two terms integrate to $1/(s-1) - 1/s$ after the factor $1/2$. Substitution $y = 1/x$ in the third gives the second term in the integral of (1.15).

The exponential bound in Lemma 1.5.2 makes that integral entire: on every compact set, powers of x and any fixed powers of $\log x$ are bounded by an integrable exponential majorant. The pole, residue, and symmetry assertions now follow directly. In particular

$$X(s) = \frac{1}{2} + \frac{s(s-1)}{4} \int_1^\infty (\theta(x) - 1) \left(x^{s/2} + x^{(1-s)/2} \right) \frac{dx}{x}.$$

For $|s| \leq R$, the integral is bounded by a constant times $\int_1^\infty e^{-\pi x} x^{(R+1)/2} dx$. If $r = (R+1)/2$, then $x^r e^{-\pi x/2} \leq (2r/(\pi e))^r$, by maximizing its logarithm. The remaining exponential is integrable. This proves (1.17), including the polynomial prefactor.

For $\operatorname{Re} s > 1$, the gamma factor and the prime product are nonzero, so X is nonzero there. Reflection excludes zeros for $\operatorname{Re} s < 0$. $\square$

The continuation of the original Dirichlet series is now the specific meromorphic function $\zeta(s) = \pi^{s/2} E(s/2) \mathcal{L}(s)$. It has a simple pole at one. At zero the zero of $E(s/2)$ cancels the pole of $\mathcal{L}$ and gives $\zeta(0) = -1/2$. At each negative even integer, $E(s/2)$ has a simple zero and $\mathcal{L}(s) = \mathcal{L}(1-s) \neq 0$. These are its trivial zeros. Its other zeros are precisely the zeros of X , with the same multiplicities. We call them the nontrivial zeros. No assertion about their real parts inside the closed strip has been used.

The closed strip can be sharpened by a positive trigonometric identity. The argument uses the pole of ζ at one and holomorphy at every other point of the line $\operatorname{Re} s = 1$.

LEMMA 1.5.4 (*The pole and the boundary germs*). The continuation $\zeta(s) = \pi^{s/2} E(s/2) \mathcal{L}(s)$ is holomorphic on $\mathbb{C} \setminus \{1\}$ and has a simple pole of residue one at $s = 1$. For every nonzero real t , it is holomorphic on neighborhoods of $1 + it$ and $1 + 2it$.

Proof. Theorem 1.5.3 makes $\mathcal{L}$ holomorphic away from zero and one. Lemma 1.4.2 makes E entire, with $E(z) = z + O(z^2)$ at zero. Thus $\pi^{s/2} E(s/2) \mathcal{L}(s) = -1/2 + O(s)$ there, and its singularity at zero is removable. The Gaussian integral gives

$$\Gamma(1/2) = 2 \int_0^\infty e^{-u^2} du = \sqrt{\pi}.$$

Consequently $\pi^{1/2} E(1/2) = 1$. The residue of $\mathcal{L}$ at one is one, so

$$\zeta(s) = \frac{1}{s-1} + H(s) \tag{1.18}$$

on a neighborhood of one, with H holomorphic. The asserted local holomorphy follows because $t \neq 0$ places both $1 + it$ and $1 + 2it$ away from the only pole. $\square$

LEMMA 1.5.5 (*A positive prime sum*). For every real $\sigma > 1$ and real t ,

$$\zeta(\sigma)^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1. \quad (1.19)$$

Proof. On $\operatorname{Re} s > 1$, define the holomorphic function

$$B(s) = \sum_p \sum_{r \geq 1} \frac{p^{-rs}}{r}.$$

Here p runs over positive primes and $p^{-rs} = e^{-rs \log p}$. For every $\delta > 0$, its series converges absolutely and uniformly on $\operatorname{Re} s \geq 1 + \delta$, since

$$\sum_p \sum_{r \geq 1} \frac{p^{-r(1+\delta)}}{r} \leq \frac{1}{1 - 2^{-1-\delta}} \sum_{n \geq 2} n^{-1-\delta} < \infty.$$

Lemma 1.4.1 gives $e^{B(s)} = \zeta(s)$. Taking absolute values yields $\operatorname{Re} B(s) = \log |\zeta(s)|$, with the real logarithm of a positive number. This operation needs no choice of a logarithm at a boundary point. Absolute convergence permits the three series to be combined. The identity

$$3 + 4 \cos \theta + \cos(2\theta) = 2(1 + \cos \theta)^2 \geq 0 \quad (\theta \in \mathbb{R})$$

therefore gives

$$\begin{aligned} & 3 \log \zeta(\sigma) + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| \\ &= \sum_p \sum_{r \geq 1} \frac{p^{-r\sigma}}{r} (3 + 4 \cos(rt \log p) + \cos(2rt \log p)) \geq 0. \end{aligned}$$

The Dirichlet series makes $\zeta(\sigma) > 0$. Exponentiation proves (1.19). $\square$

The exponents in (1.19) decide the boundary case. A simple pole contributes order -3 , while even a simple zero in the middle factor contributes order 4. The third factor is bounded at the limiting point whenever $t \neq 0$.

THEOREM 1.5.6 (*Nonvanishing on the boundary and the strict strip*). For every real $t \neq 0$, one has $\zeta(1 + it) \neq 0$. At $s = 1$, the function ζ has a simple pole of residue one. Every zero ρ of X satisfies

$$0 < \operatorname{Re} \rho < 1, \quad z_\rho = -i(\rho - 1/2), \quad |\operatorname{Im} z_\rho| < \frac{1}{2}. \quad (1.20)$$

The zeros of X are exactly the nontrivial zeros of ζ , with the same multiplicities. Reflection and complex conjugation preserve these multiplicities.

Proof. Fix $t \in \mathbb{R} \setminus \{0\}$ and suppose $\zeta(1 + it) = 0$. By Lemma 1.5.4, this is a zero of a holomorphic function. It has some finite order $m \geq 1$. Indeed, a zero of infinite order would make the function vanish on a disc, contrary to its nonzero Euler product in the same disc's intersection with $\operatorname{Re} s > 1$. Taylor expansion gives a holomorphic function a , with $a(0) \neq 0$, such that

$$\zeta(1 + it + w) = w^m a(w)$$

for sufficiently small complex w . For positive real h tending to zero, the pole expansion and local holomorphy at $1 + 2it$ give constants $C_0, C_1, C_2 > 0$ such that

$$0 < \zeta(1+h) \leq C_0 h^{-1}, \quad |\zeta(1+h+it)| \leq C_1 h^m, \quad |\zeta(1+h+2it)| \leq C_2.$$

The constants can depend on t and on the chosen neighborhoods. In particular, boundedness of the third factor allows a zero at $1 + 2it$. Lemma 1.5.5 now implies

$$1 \leq \zeta(1+h)^3 |\zeta(1+h+it)|^4 |\zeta(1+h+2it)| \leq C_0^3 C_1^4 C_2 h^{4m-3} \rightarrow 0.$$

This contradiction proves nonvanishing for $t \neq 0$. When $t = 0$, Lemma 1.5.4 gives the pole directly. Thus the preceding limiting argument never treats the pole as a holomorphic value of ζ .

For $s = 1 + it$ with $t \neq 0$, the factors $s, s-1$, and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$ are finite and nonzero. The nonvanishing just proved therefore gives $X(1+it) \neq 0$. The value $X(1) = 1/2$ handles $t = 0$. On the other boundary, the exact reflection identity gives $X(it) = X(1-it) \neq 0$ for every real t , including $X(0) = 1/2$. The closed-strip statement in Theorem 1.5.3 now yields $0 < \operatorname{Re} \rho < 1$ for every zero ρ of X .

On this open strip, $s(s-1)\Gamma_{\mathbb{R}}(s)/2$ is holomorphic and nonzero. Multiplication by it preserves each zero order, so the zeros of X and ζ there have the same multiplicities. At $s = 0$, one has $\zeta(0) = -1/2$. For $\operatorname{Re} s < 0$, reflection gives $\mathcal{L}(s) = \mathcal{L}(1-s) \neq 0$. The reciprocal gamma factor has simple zeros exactly at $s = 0, -2, -4, \dots$. It follows that the zeros of ζ in $\operatorname{Re} s < 0$ are exactly the simple zeros at $s = -2, -4, \dots$. These are the trivial zeros. On $\operatorname{Re} s = 0$ with $s \neq 0$, both $E(s/2)$ and $\mathcal{L}(s)$ are nonzero. The Euler product excludes zeros for $\operatorname{Re} s > 1$. This proves the stated classification of all other zeros.

If $\rho = \beta + i\gamma$ with $\beta, \gamma \in \mathbb{R}$, then

$$z_\rho = \gamma + i(1/2 - \beta).$$

Thus $0 < \beta < 1$ is exactly the strict bound in (1.20). The affine map $s \mapsto -i(s - 1/2)$ has nonzero derivative. Consequently a zero of order m at $s = \rho$ is a zero of order m of $z \mapsto X(1/2 + iz)$ at $z = z_\rho$. Finally the identities $X(s) = X(1-s)$ and $X(\bar{s}) = \overline{X(s)}$ preserve orders in the local power series. In centered variables their images are

$$z_{1-\rho} = -z_\rho, \quad z_{\bar{\rho}} = -\overline{z_\rho}, \quad z_{1-\bar{\rho}} = \overline{z_\rho}.$$

All three parameters therefore occur with the same multiplicity as z_ρ , with coincident parameters counted only according to their actual zero order. $\square$

Lemma 1.3.1 supplies exponential decay for every test in $\mathcal{A}$. The same centered coordinate specifies where the Weil transform is evaluated. Suppose $a > 1/2$ and $g : \mathbb{R} \rightarrow \mathbb{C}$ is continuous with $|g(u)| \leq C e^{-a|u|}$. Define

$$\widehat{g}(z) = \int_{\mathbb{R}} g(u) e^{izu} du, \quad F_g(s) = \widehat{g}(-i(s - 1/2)) = \int_{\mathbb{R}} g(u) e^{(s-1/2)u} du.$$

For each $0 \leq b < a$ and integer $k \geq 0$, the absolute value of the k th derivative integrand is bounded on $|\operatorname{Im} z| \leq b$ by

$$C|u|^k e^{-(a-b)|u|}.$$

This is integrable on $\mathbb{R}$. Termwise differentiation therefore makes $\widehat{g}$ holomorphic on $|\operatorname{Im} z| < a$. It makes F_g holomorphic on $1/2 - a < \operatorname{Re} s < 1/2 + a$. Every nontrivial zero lies in this domain, and $F_g(\rho) = \widehat{g}(z_\rho)$ uses the same multiplicity under the invertible affine change of variable.

The strict strip permits zeros with $\operatorname{Re} \rho \neq 1/2$. The assertion that all z_ρ are real is the Riemann hypothesis. It remains an additional assertion beyond boundary nonvanishing.

COROLLARY 1.5.7 (*Zero count and absolute test sums*). The number of nontrivial zeros with $|\rho| \leq R$, counted with multiplicity, is $O(R \log(R+2))$. For every $g \in \mathcal{A}$, the series $\sum_{\rho} |F_g(\rho)|$ converges.

Proof. Apply Lemma 1.2.3 to X , using $X(o) = 1/2$. Since $0 \leq \operatorname{Re} \rho \leq 1$, Lemma 1.3.1 gives $|F_g(\rho)| \leq C_N(1 + |\operatorname{Im} \rho|)^{-N}$ for every N . The zeros in $2^j \leq |\operatorname{Im} \rho| < 2^{j+1}$ contribute at most $C(j+1)2^{j(1-N)}$. Taking $N = 3$ gives a convergent series. The bounded region has only finitely many zeros. $\square$

1.6 HEIGHTS AT WHICH THE HORIZONTAL INTEGRALS VANISH

Write $H(s) = \mathcal{L}'(s)/\mathcal{L}(s)$. It has residue equal to the multiplicity at each nontrivial zero. Its residue at each of zero and one is -1 . These signs follow by differentiating the logarithm of a local factor $(s - s_o)^m$ or $(s - s_o)^{-1}$. The identity

$$H(s) = \frac{X'(s)}{X(s)} - \frac{1}{s} - \frac{1}{s-1}, \quad H(1-s) = -H(s) \quad (1.21)$$

will control both the poles and the left vertical line.

LEMMA 1.6.1 (*Admissible rectangle heights*). There are heights $T_m \in [m, m+1]$, for all sufficiently large integers m , such that no zero lies on either horizontal segment $-1/2 \leq \operatorname{Re} s \leq 3/2$, $\operatorname{Im} s = \pm T_m$, and

$$\sup_{-1/2 \leq \sigma \leq 3/2} |H(\sigma \pm iT_m)| = O(T_m^2 \log^2(T_m + 2)). \quad (1.22)$$

For $g \in \mathcal{A}_a$ and $1 < c < \min(a + 1/2, 3/2)$, both horizontal integrals of $H(s)F_g(s)$ between the lines $1-c$ and c tend to zero at these heights.

Proof. Let N_m count all zeros in $|s| \leq m+8$, with multiplicity, and set $\delta_m = 1/(8(N_m + 1))$. Remove from $[m, m+1]$ the intervals of radius δ_m about the absolute values of their imaginary parts. The removed length is at most $2\delta_m N_m < 1/4$, so a remaining T_m exists. Zeros outside this disc have $|\operatorname{Im} \rho| > m+1+\delta_m$ for large m , because $0 \leq \operatorname{Re} \rho \leq 1$. Thus both heights have distance at least δ_m from every zero ordinate. Corollary 1.5.7 gives $\delta_m^{-1} = O(m \log(m+2))$.

Fix $T = T_m$ and center a disc at $s_o = 2 + iT$. By (1.17), $|X(s_o + z)| \leq \exp(C(T+8) \log(T+8))$ for $|z| \leq 6$. At its center, the prime-product lower bound and (1.11) give

$$|X(2 + iT)| = \frac{|(2 + iT)(1 + iT)|}{2\pi} |\Gamma(1 + iT/2)| |\zeta'(2 + iT)| \geq \exp(-C(T+8) \log(T+8)).$$

Choose $R \in [5, 6]$ with no zero on the circle. The number of zeros in it is $O(T \log(T+2))$ by the global count. Every point of the desired horizontal segment has $|s - s_o| \leq 5/2$, and its distance from every zero is at least δ_m . Lemma 1.2.4 now gives

$$|X'(s)/X(s)| \leq CT \log(T+2)(1 + \delta_m^{-1}) = O(T^2 \log^2(T+2)).$$

The two rational terms in (1.21) are bounded at these heights. Conjugation gives the same estimate on the bottom segment.

The chosen vertical strip lies strictly inside $1/2 - a < \operatorname{Re} s < 1/2 + a$. Its test transform is $O(T^{-4})$ uniformly by Lemma 1.3.1. The horizontal lengths are bounded, so their integrals are $O(T^{-2} \log^2(T+2))$, which tends to zero. $\square$

The proof uses a polynomial bound, which suffices for every test in $\mathcal{A}$. It does not require a sharper local zero count, a product over all zeros, or an asymptotic formula for the number of zeros.

1.7 THE NORMALIZED EXPLICIT FORMULA

THEOREM 1.7.1 (*The full arithmetic functional*). For every $g \in \mathcal{A}$, equation (1.4) holds, with all sums and integrals absolutely convergent. Its right side defines a linear functional $L : \mathcal{A} \rightarrow \mathbb{C}$ satisfying

$$L(g^*) = \overline{L(g)}, \quad L(g) = \sum_{\rho} \widehat{g}(-i(\rho - 1/2)). \quad (1.23)$$

Proof. Choose $g \in \mathcal{A}_a$ and $1 < c < \min(a + 1/2, 3/2)$. The prime terms converge because Lemma 1.3.1 gives

$$\sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (|g(\log n)| + |g(-\log n)|) \leq C \sum_{n \geq 2} (\log n) n^{-a-1/2} < \infty.$$

The gamma integral converges by (1.12) and (1.5). The polar values are inside the transform strip, and the zero sum converges by Corollary 1.5.7.

Integrate $H(s)F_g(s)$ around the rectangle with vertical sides $1 - c, c$ and heights $\pm T_m$, counterclockwise. Its right side is traversed upward. There are no poles on the vertical sides, since the zeros lie in the closed critical strip. The residue formula gives the zero sum inside the rectangle minus $F_g(0) + F_g(1)$. Lemma 1.6.1 eliminates the horizontal integrals. On $\text{Re } s = c$, equations (1.9) and (1.12) give $H(c + it) = O(\log(2 + |t|))$. Reflection gives the same bound on $1 - c$, so both vertical integrals converge absolutely. In the left integral, substitute $s = 1 - w$ and use $H(1 - w) = -H(w)$. Both vertical-line integrals in the resulting expression are oriented upward. The outcome is

$$\sum_{\rho} F_g(\rho) - F_g(0) - F_g(1) = \frac{1}{2\pi i} \int_{\text{Re } s=c} H(s)(F_g(s) + F_g(1-s)) ds. \quad (1.24)$$

On this line, split $H = H_{\Gamma} + \zeta'/\zeta$. The series for ζ'/ζ can pass through the integral because $\sum \Lambda_{\text{vM}}(n)n^{-c} < \infty$ and both test factors are integrable on the vertical line. For the first factor, Fourier inversion applied to $u \mapsto e^{(c-1/2)u} g(u)$ gives

$$\frac{1}{2\pi i} \int_{\text{Re } s=c} n^{-s} F_g(s) ds = n^{-c} e^{(c-1/2)\log n} g(\log n) = n^{-1/2} g(\log n). \quad (1.25)$$

This weighted function has exponential decay with all derivatives, since $c - 1/2 < a$. The proof of Lemma 1.3.2 therefore applies to it. For $F_g(1 - s)$, use $u \mapsto e^{(c-1/2)u} g(-u)$ instead. It gives $n^{-1/2} g(-\log n)$. The minus sign in (1.9) proves exactly the prime-power term in (1.4).

It remains to evaluate the gamma contribution. The function H_{Γ} is holomorphic for $\text{Re } s > 0$. Shift each gamma integral from c to $1/2$. There is no pole between these lines, and the logarithmic bound (1.12) times the rapid test decay eliminates the horizontal sides. Set $s = 1/2 + it$. The two factors are $\widehat{g}(t)$ and $\widehat{g}(-t)$. Changing t to $-t$ in the second term gives

$$\begin{aligned} & \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) (H_{\Gamma}(1/2 + it) + H_{\Gamma}(1/2 - it)) dt \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) [\text{Re } \psi(1/4 + it/2) - \log \pi] dt. \end{aligned}$$

Here conjugation of Γ follows from its defining real integral and analytic continuation. Together with (1.24), this proves (1.4).

Finally the gamma bracket is real and even. For real t , (1.8) gives $\widehat{g}^*(t) = \overline{\widehat{g}(t)}$. The involution interchanges the two polar values and the two values at $\pm \log n$, followed by conjugation. Thus all three local terms obey the Hermitian identity in (1.23). $\square$

The proof explains the normalization directly. The poles of $\mathcal{L}$ each contribute residue -1 to its logarithmic derivative, giving the two positive polar values after rearrangement. The logarithmic derivative of each Euler factor gives the negative prime term. The reflection combines the two gamma factors at $1/2 \pm it$. These operations preserve the original complex test function throughout.

1.8 GAUSSIAN TESTS AND THEIR BOUNDARY TERMS

For g_A from Lemma 1.3.2, substitution in the formula gives

$$\begin{aligned} \sum_{\rho} e^{A(\rho-1/2)^2} &= 2e^{A/4} + \frac{1}{2\pi} \int_{\mathbb{R}} e^{-At^2} [\operatorname{Re} \psi(1/4 + it/2) - \log \pi] dt \\ &\quad - \frac{1}{\sqrt{\pi A}} \sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} e^{-(\log n)^2/(4A)}. \end{aligned} \quad (1.26)$$

If every nontrivial zero has real part $1/2$, each term in this Gaussian zero sum is positive. The identity itself is unconditional.

PROPOSITION 1.8.1 (*Polynomial–Gaussian compatibility*). Let $A > 0$, $\tau \in \mathbb{R}$, and $P \in \mathbb{C}[w]$. The function

$$f(u) = e^{-i\tau u} P(i\partial_u) g_A(u) \quad (1.27)$$

belongs to $\mathcal{A}_a$ for every $a > 0$, and

$$\widehat{f}(z) = P(z - \tau) e^{-A(z-\tau)^2}. \quad (1.28)$$

For every fixed $b > 0$ and $z = t + iy$ with $|y| \leq b$,

$$|\widehat{f}(z)| \leq C_{P,A,b,\tau} (1 + |t|)^{\deg P} e^{-A(t-\tau)^2 + Ab^2}. \quad (1.29)$$

Consequently every real integration-by-parts boundary value and every horizontal contour term used in Theorem 1.7.1 vanishes for f and $f^* \star f$.

Proof. Each derivative in (1.27) is a polynomial in u times $e^{-u^2/(4A)}$, multiplied by $e^{-i\tau u}$. This description holds after any further derivative. Multiplication by $e^{a|u|}$ still leaves an integrable function tending to zero at both infinities, for every $a > 0$. Integration by parts gives $i\partial_u \widehat{h}(z) = z\widehat{h}(z)$, and modulation gives $e^{-i\tau u} \widehat{h}(z) = \widehat{h}(z - \tau)$. Together these prove (1.28). The exact Gaussian modulus is $|e^{-A(t-\tau+iy)^2}| = e^{-A(t-\tau)^2 + Ay^2}$, which proves (1.29). This bound times (1.22) tends to zero at both heights. The convolution square lies in every $\mathcal{A}_a$ by Proposition 1.3.3; its transform is the product in (1.8), with Gaussian decay of twice the exponent. Thus it has the same boundary properties. $\square$

Example 1.8.2 (*The first polynomial and a missing conjugation*). For $P(w) = w$, the function and its transform are

$$\begin{aligned} f(u) &= -iu e^{-i\tau u} g_A(u)/(2A), \\ \widehat{f}(z) &= (z - \tau) e^{-A(z-\tau)^2}. \end{aligned}$$

Its convolution square has transform

$$\widehat{f^* \star f}(z) = (z - \tau)^2 e^{-2A(z-\tau)^2}.$$

At $z = \tau + ib$, this value is $-b^2 e^{2Ab^2}$. The correct analytic expression is $\overline{\widehat{f}(\bar{z})} \widehat{f}(z)$. Replacing it by $|\widehat{f}(z)|^2$ would erase the negative sign and would not give a holomorphic function of z .

1.9 A SEPARATE POSITIVITY ARGUMENT

The explicit formula compares linear functionals. Positivity concerns its value on every convolution square and requires a different argument. Define the Riemann hypothesis to mean that every nontrivial zero ρ satisfies $\operatorname{Re} \rho = 1/2$.

THEOREM 1.9.1 (*Positivity on the exact test algebra*). The Riemann hypothesis is equivalent to

$$L(f^* \star f) \geq 0 \quad \text{for every } f \in \mathcal{A}. \quad (1.30)$$

Under the Riemann hypothesis, writing $\rho = 1/2 + i\gamma_\rho$ gives $L(f^* \star f) = \sum_\rho |\widehat{f}(\gamma_\rho)|^2$. If there is a zero off the critical line, a function of the form (1.27) has strictly negative convolution-square value.

Proof. Put $z_\rho = -i(\rho - 1/2)$. Theorem 1.5.3 gives $|\operatorname{Im} z_\rho| \leq 1/2$. Its reflection and conjugation symmetries show that z_ρ and $\bar{z}_\rho$ occur with equal multiplicity. Under the Riemann hypothesis all these parameters are real. Then Theorem 1.7.1 and (1.8) give the asserted nonnegative sum.

For the converse, suppose $z_0 = \tau + ib$ occurs, where $b \neq 0$. Its conjugate has the same multiplicity m . Center the parameter by $w = z - \tau$. There are finitely many distinct centered zero parameters with $|\operatorname{Re} w| \leq 2$. This follows from the global zero count and the bound on their imaginary parts. Form the monic polynomial Q whose roots are these distinct points except ib and $-ib$. If there are none, set $Q = 1$. This root set is invariant under conjugation, so Q has real coefficients and $Q(ib)Q(-ib) = |Q(ib)|^2 \neq 0$. Set

$$P(w) = wQ(w)Q(-w), \quad \widehat{f}_A(z) = P(z - \tau)e^{-A(z-\tau)^2}, \quad A \geq 1.$$

The polynomial P is real and odd. Proposition 1.8.1 constructs $f_A \in \mathcal{A}$ exactly. Since τ and the coefficients are real, $\widehat{f}_A(\bar{z}) = \widehat{f}_A(z)$. The two selected zero parameters therefore contribute

$$-2mb^2|Q(ib)|^4 e^{2Ab^2} \quad (1.31)$$

to $L(f_A^* \star f_A)$. All other centered zero parameters with $|\operatorname{Re} w| \leq 2$ give zero because Q vanishes there.

For a remaining parameter $w = x + iy$, one has $|x| > 2$, $|y| \leq 1/2$, and $D = x^2 - y^2 \geq 15/4$. For $A \geq 1$, $2AD \geq 15A/4 + (x^2 - 1/4)$. Thus the absolute value of the remaining sum is bounded by

$$e^{-15A/4} \sum_{|\operatorname{Re} w| > 2} |P(w)|^2 e^{-(x^2 - 1/4)}. \quad (1.32)$$

The final series is finite by the polynomial growth of P and the zero count. It is independent of A . Expression (1.31) tends to minus infinity while (1.32) tends to zero. The Hermitian identity in (1.23) makes the full value real, so it is strictly negative for sufficiently large A . This contradicts (1.30) and proves the converse. $\square$

The conclusion is a criterion, not a proof of the Riemann hypothesis. The transform estimates, the continuation, and the contour formula were proved before invoking any sign condition. Their exact comparison identifies the remaining arithmetic question: whether the full local expression is nonnegative on every convolution square. The negative separator describes what any off-line zero would force.

1.10 THE FORM ON THE NONUNITAL TEST ALGEBRA

For the representation theory, write $\mathcal{A}_\zeta = \mathcal{A}$ and $L_\zeta = L$. These are identical sets, algebra operations, and functional values. This identification takes no completion. Membership in $\mathcal{A}$ means that there is $a > 1/2$ with every $p_{a,k}$ finite. Setting $\epsilon = a - 1/2$ says precisely that $e^{(1/2+\epsilon)|u|} g^{(k)}(u)$ is integrable for every k . The involution is the reflection and conjugation in (1.2); its square transform is the analytic product (1.8).

The operations on this algebra are

$$(f \star g)(u) = \int_{\mathbb{R}} f(v) g(u - v) dv, \quad f^*(u) = \overline{f(-u)}. \quad (1.33)$$

On each common holomorphic strip their transforms obey

$$\widehat{f \star g}(z) = \widehat{f}(z) \widehat{g}(z), \quad \widehat{f^*}(z) = \overline{\widehat{f}(\bar{z})}. \quad (1.34)$$

These are the operations and identities already proved in (1.2) and Proposition 1.3.3. For real u , the same Fourier convention and Lemma 1.3.2 give

$$\widehat{f}(z) = \int_{\mathbb{R}} f(u) e^{izu} du, \quad f(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t) e^{-itu} dt. \quad (1.35)$$

In particular, the product identity on the real Fourier axis is

$$\widehat{f^* \star f}(t) = |\widehat{f}(t)|^2 \quad (t \in \mathbb{R}). \quad (1.36)$$

The derivative of the gamma normalization in (1.13) reads

$$2 \operatorname{Re} \frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}(1/2 + it) = \operatorname{Re} \psi(1/4 + it/2) - \log \pi. \quad (1.37)$$

The multiplicative variable $x = e^u$ gives the identity

$$\widehat{g}(-i(s - 1/2)) = \int_0^\infty g(\log x) x^{s-1/2} \frac{dx}{x}, \quad 1/2 - a < \operatorname{Re} s < 1/2 + a \quad (1.38)$$

for $g \in \mathcal{A}_a$. The centered Mellin convention gives $F_g(0) = \widehat{g}(i/2)$ and $F_g(1) = \widehat{g}(-i/2)$. Thus the functional constructed in Theorem 1.7.1 is exactly

$$\begin{aligned} L_\zeta(g) &= \widehat{g}(i/2) + \widehat{g}(-i/2) \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) [\operatorname{Re} \psi(1/4 + it/2) - \log \pi] dt \\ &- \sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (g(\log n) + g(-\log n)). \end{aligned} \quad (1.39)$$

All sums and integrals converge absolutely on this algebra. Theorem 1.7.1 also gives the exact zero expression

$$L_\zeta(g) = \sum_{\rho} \widehat{g}(-i(\rho - 1/2)). \quad (1.40)$$

A sesquilinear form is linear in its first argument and conjugate-linear in its second. It is Hermitian when reversing the arguments conjugates the value, and positive when its diagonal values are nonnegative real numbers. Define the sesquilinear form by

$$B_\zeta(f, g) = L_\zeta(g^* \star f).$$

The identity

$$L_\zeta(g^*) = \overline{L_\zeta(g)} \quad (1.41)$$

from Theorem 1.7.1 makes B_ζ Hermitian. Theorem 1.9.1 states

$$\text{RH} \iff L_\zeta(f^* \star f) \geq 0 \quad \text{for every } f \in \mathcal{A}_\zeta. \quad (1.42)$$

Under RH, the same theorem gives

$$L_\zeta(f^* \star f) = \sum_{\rho} |\widehat{f}(\gamma_\rho)|^2, \quad \rho = 1/2 + i\gamma_\rho. \quad (1.43)$$

The right side explains positivity conditional on RH. Both the reality of the parameters γ_ρ and their occurrence in this expression use the stated zero placement. They do not construct positivity from the arithmetic local terms independently.

Theorem 1.5.6 proves $0 < \text{Re } \rho < 1$ and $|\text{Im } z_\rho| < 1/2$, with the same zero multiplicities in both coordinates. The closed bounds used in the contour and separator follow from these strict bounds. Nonreal z_ρ are still permitted, so boundary nonvanishing supplies no independent positivity of B_ζ .

1.11 POSITIVE FUNCTIONALS AND THEIR REPRESENTATIONS

A $*$ -homomorphism preserves addition, complex scalar multiplication, products, and this involution; no preservation of a unit is imposed here. For a positive form, expansion of the diagonal at $x + \lambda y$ gives Hermitian symmetry and

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

Indeed, when $\langle y, y \rangle > 0$, minimize that quadratic expression in λ ; if it is zero, arbitrarily large multiples force the cross term to vanish. A positive definite form gives a norm $\|x\| = \sqrt{\langle x, x \rangle}$. Its space is a pre-Hilbert space. Completing Cauchy sequences modulo sequences tending to zero gives a Hilbert space; continuity extends the inner product to that completion. An isometry preserves this inner product, and a surjective isometry is unitary. An operator is bounded if $\|Sx\| \leq C\|x\|$ on its domain; the least such constant is its operator norm.

Orthogonality means vanishing of the inner product. For a linear subspace M of a Hilbert space H , write

$$M^\perp = \{v \in H : \langle v, m \rangle = 0 \text{ for every } m \in M\}.$$

Closed subspaces have the following decomposition, which will distinguish a dense operator range from a proper one.

LEMMA 1.11.1 (*Projection onto a closed subspace*). If $M \subset H$ is a closed linear subspace, every $v \in H$ has a unique expression $v = m + n$ with $m \in M$ and $n \in M^\perp$. In particular, $M^\perp = \{0\}$ implies $M = H$.

Proof. Put $d^2 = \inf_{m \in M} \|v - m\|^2$ and choose $m_j \in M$ with $\|v - m_j\|^2 \rightarrow d^2$. Expansion of inner products gives the parallelogram identity and hence

$$\|m_j - m_k\|^2 = 2\|v - m_j\|^2 + 2\|v - m_k\|^2 - 4\left\|v - \frac{m_j + m_k}{2}\right\|^2 \leq 2\|v - m_j\|^2 + 2\|v - m_k\|^2 - 4d^2.$$

Thus (m_j) is Cauchy. Completeness and closedness give a limit $m \in M$ attaining the infimum. For $y \in M$, the real function $t \mapsto \|v - m - ty\|^2$ is minimized at zero, so its linear coefficient gives $\operatorname{Re}\langle v - m, y \rangle = 0$. Replacing y by iy also makes the imaginary part zero. Therefore $n = v - m$ lies in $M^\perp$. The intersection $M \cap M^\perp$ is zero, since an element of that intersection has squared norm zero. This proves uniqueness and the last assertion. $\square$

An operator on a Hilbert space H means a linear map $S : D \rightarrow H$ on a specified dense linear subspace. Its graph is $\{(x, Sx) : x \in D\}$. It is closed when its graph is closed in $H \oplus H$, and closable when the graph closure is again a graph. Equivalently, $x_n \rightarrow 0$ and $Sx_n \rightarrow y$ must imply $y = 0$; this equivalence follows by subtracting two possible limiting graph values at the same vector. The graph norm is $(\|x\|^2 + \|Sx\|^2)^{1/2}$. A core for a closable operator is a subspace of its domain dense in this norm. The adjoint has domain

$$\mathcal{D}(S^*) = \{y \in H : \text{there is } z \in H \text{ with } \langle Sx, y \rangle = \langle x, z \rangle \text{ for all } x \in D\}, \quad S^*y = z.$$

Density makes z unique. Passing to limits in this identity shows that the adjoint is closed. If this adjoint domain is dense, the preceding sequence test proves closability: a proposed y pairs to zero with every adjoint-domain vector. A symmetric operator satisfies $S \subseteq S^*$; a self-adjoint operator satisfies equality, including its domain.

A representation on D is an algebra homomorphism into linear maps $D \rightarrow D$. It is a $*$ -representation when $\langle \pi(a)x, y \rangle = \langle x, \pi(a^*)y \rangle$ for $x, y \in D$. Thus the algebraic domain must be invariant before the adjoint identity can be used in repeated products. We now construct such a representation from a positive functional; this is the nonunital GNS construction.

Let $\mathfrak{A}$ be a complex $*$ -algebra, with no norm or unit assumed, and let $W : \mathfrak{A} \rightarrow \mathbb{C}$ be positive:

$$W(a^*a) \geq 0 \quad (a \in \mathfrak{A}).$$

THEOREM I.II.2 (*Nonunital algebraic GNS*). Put

$$\langle a, b \rangle_W = W(b^*a), \quad \mathcal{N}_W = \{a : W(a^*a) = 0\}.$$

Then $\mathcal{N}_W$ is a left ideal, meaning a linear subspace preserved by left multiplication by every algebra element, and

$$D_W = \mathfrak{A}/\mathcal{N}_W, \quad \pi_W(a)[b] = [ab], \quad (1.44)$$

is a $*$ -representation on a pre-Hilbert space. Regard D_W as a common invariant dense domain in its Hilbert completion $\mathcal{H}_W$. For every $a \in \mathfrak{A}$,

$$\pi_W(a^*) \subseteq \pi_W(a)^*, \quad (1.45)$$

so $\pi_W(a)$ is closable. It extends boundedly to $\mathcal{H}_W$ if and only if there is a constant C_a such that

$$W(b^*a^*ab) \leq C_a^2 W(b^*b) \quad (b \in \mathfrak{A}). \quad (1.46)$$

The least such C_a is the norm of the extension.

Proof. Positivity and polarization make $\langle \cdot, \cdot \rangle_W$ Hermitian and give Cauchy–Schwarz. If $b \in \mathcal{N}_W$, then

$$W((ab)^*ab) = \langle a^*ab, b \rangle_W = 0,$$

so $\mathcal{N}_W$ is a left ideal and (1.44) is well defined. For classes in D_W ,

$$\langle \pi_W(a)[b], [c] \rangle_W = W(c^*ab) = \langle [b], \pi_W(a^*)[c] \rangle_W.$$

This proves (1.45). The adjoint domain contains the dense subspace D_W , so $\pi_W(a)$ is closable. Finally,

$$\|\pi_W(a)[b]\|^2 = W(b^*a^*ab),$$

so the bounded-operator criterion is exactly (1.46). $\square$

For $a = a^*$, the operator $\pi_W(a)$ is symmetric and closable. The closure need not be self-adjoint or admit a self-adjoint extension. The obstruction can be checked directly. For a closed symmetric operator S , expansion gives $\|(S \pm i)x\|^2 = \|Sx\|^2 + \|x\|^2$. Consequently both ranges are closed, and

$$V : (S + i)x \mapsto (S - i)x$$

is an isometry from $\text{ran}(S + i)$ onto $\text{ran}(S - i)$. Their orthogonal complements are respectively $\ker(S^* - i)$ and $\ker(S^* + i)$, by the definition of the adjoint. If a self-adjoint T extends S , each $T \pm i$ has closed dense range: its orthogonal complement is $\ker(T \mp i) = 0$ by the norm identity. Lemma 1.11.1 therefore makes each range all of H . Thus $(T - i)(T + i)^{-1}$ is a unitary extension of V . Its restrictions to the two range complements must be unitary too, so these complements must have the same dimension.

For example,

$$P = -i \frac{d}{dx}, \quad \mathcal{D}(P) = C_c^\infty(0, \infty) \subset L^2(0, \infty),$$

Here $C_c^\infty(0, \infty)$ consists of smooth functions with compact support inside the half-line. It is dense in L^2 : truncate the support and size of an L^2 function, approximate by step functions, and replace interval endpoints by smooth ramps on sets of arbitrarily small length. The operator is symmetric by integration by parts, and hence closable by the dense adjoint-domain criterion. The adjoint equations, tested against compactly supported smooth functions, say weakly that $u' = -u$ and $u' = u$. Multiplication by the integrating factors e^x and e^{-x} gives a weak zero derivative. A locally integrable function with weak zero derivative is constant: every compact test function of integral zero is the derivative of another compact test function, so the resulting distribution depends only on the integral of its test. The solutions are therefore ce^{-x} and ce^x , respectively. Only the first belongs to $L^2(0, \infty)$. Passing to the closure does not change the adjoint, since its defining identity passes through graph limits. Its two range complements thus have dimensions one and zero, which excludes a self-adjoint extension.

This operator is itself a GNS multiplier. Put $D = \mathcal{D}(P)$ and use on $L^2(0, \infty)$ the inner product linear in its first argument. For $f, g \in D$, define the rank-one operator

$$R_{f,g}x = \langle x, g \rangle f \quad (x \in D).$$

Let $\mathfrak{A}_D$ be the unital algebra of operators $D \rightarrow D$ generated by P and all $R_{f,g}$. Every generator preserves D ; integration by parts and the inner product identity give adjoints on D by $I^* = I$, $P^* = P$, and $R_{f,g}^* = R_{g,f}$. For a finite linear combination of words, conjugate the coefficients and reverse each word with these adjoints. The resulting operator B^* satisfies $\langle Bx, y \rangle = \langle x, B^*y \rangle$ for $x, y \in D$. If two

expressions define the same operator B , this identity and density of D make their adjoint operators equal. Thus the involution is well-defined on $\mathfrak{A}_D$, and is the restriction of the Hilbert adjoint to D . It makes $\mathfrak{A}_D$ a $*$ -algebra with invariant domain D .

Choose $\Omega \in D$ with $\|\Omega\| = 1$ and put $W(B) = \langle B\Omega, \Omega \rangle$ for $B \in \mathfrak{A}_D$. Then

$$W(B^*B) = \|B\Omega\|^2, \quad \mathcal{N}_W = \{B : B\Omega = 0\}.$$

The map $U_o : [B] \mapsto B\Omega$ from its GNS quotient to D is an isometry, since $W(C^*B) = \langle B\Omega, C\Omega \rangle$. It is onto: $R_{f,\Omega}\Omega = f$ for every $f \in D$. Its Hilbert completion is therefore all of $L^2(o, \infty)$, and

$$U_o \pi_W(C)[B] = CB\Omega = CU_o[B] \quad (B, C \in \mathfrak{A}_D).$$

In particular, the symmetric algebra element P acts in this GNS representation as exactly the operator P with domain D . The domain is invariant under the entire algebra and is a core for $\bar{P}$ by the definition of graph closure. The preceding range-complement calculation therefore gives a symmetric GNS multiplier whose closure has no self-adjoint extension.

Products of closures require a separate common-domain construction. We give that construction in Section 1.12.

Example 1.11.3 (An unbounded positive GNS multiplier). Let $\mathfrak{A} = \mathbb{C}[x]$, let $x^* = x$, and set

$$W(p) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} p(t) e^{-t^2} dt.$$

Then W is positive, but multiplication by x is unbounded, since

$$\frac{\|\pi_W(x)[x^n]\|^2}{\|[x^n]\|^2} = \frac{\int_{\mathbb{R}} t^{2n+2} e^{-t^2} dt}{\int_{\mathbb{R}} t^{2n} e^{-t^2} dt} = n + \frac{1}{2}.$$

The last equality follows by integrating the derivative of $t^{2n+1}e^{-t^2}$; its boundary values vanish.

1.12 CLOSED FORMS AND THE COMMON INVARIANT DOMAIN

For a linear operator S on a dense subspace $D \subset H$, the quadratic form $q(x) = \|Sx\|^2$ is nonnegative. Its form norm is $(\|x\|^2 + q(x))^{1/2}$. Such a form is closable if $x_n \rightarrow 0$ in H and $q(x_n - x_m) \rightarrow 0$ imply $q(x_n) \rightarrow 0$; the implication follows from closability of S . A closed form has a complete form norm, and the closure is obtained by completing this norm and identifying the Hilbert limits. A finite form value permits one operator application. Repeated products require control of every algebra element.

For sequence examples, $\ell^2(\mathbb{N}, w)$ consists of complex sequences with $\sum_{n \geq 1} w_n |x_n|^2 < \infty$, for strictly positive weights w_n . Its inner product is $\sum_n w_n x_n \bar{y}_n$. Finite sequences, denoted $c_{oo}(\mathbb{N})$, are dense by coordinate truncation; e_n denotes the n th coordinate vector. Unspecified weights are one. The distinction between a single form and all products already occurs for a diagonal operator.

Example 1.12.1 (A maximal form domain need not be invariant). On $\ell^2(\mathbb{N})$, let $Ne_n = ne_n$, initially on $c_{oo}(\mathbb{N})$. Its closure and the closed form $q(v) = \|Nv\|^2$ have domain

$$V = \left\{ v : \sum_{n \geq 1} n^2 |v_n|^2 < \infty \right\}.$$

Coordinate convergence proves that the maximal diagonal operator is closed. Finite coordinate truncations approximate every vector in its graph norm, so they prove the asserted closure. The vector $v_n = n^{-2}$ belongs to V , but $Nv \notin V$, since $\sum n^4 |v_n|^2 = \infty$. For the polynomial algebra generated by N , the common domain is instead

$$V_\infty = \bigcap_{k \geq 0} \left\{ v : \sum_{n \geq 1} n^{2k} |v_n|^2 < \infty \right\}. \quad (1.47)$$

Multiplication by each polynomial preserves V_∞ . Truncations converge in every displayed seminorm.

Return to Theorem 1.11.2. Write $D = D_W$, $H = \mathcal{H}_W$, and $A_a = \overline{\pi_W(a)}$ for $a \in \mathfrak{A}$. The graph topology on D is generated by $\|\xi\|$ and $\|\pi_W(a)\xi\|$, for all $a \in \mathfrak{A}$. A neighborhood bounds finitely many of these seminorms, so the topology is locally convex. A representation is called closed when its domain is complete for this topology. Completion can require nets, namely families indexed by a directed set, when there is no countable family of controlling seminorms. A net is Cauchy here when it is Cauchy in every graph seminorm.

THEOREM 1.12.2 (*The exact common graph domain*). The completion of D in its graph topology identifies with

$$D_W^{\text{cl}} = \bigcap_{a \in \mathfrak{A}} \mathcal{D}(A_a) \subset H. \quad (1.48)$$

On this domain, the maps $\pi_W^{\text{cl}}(a) = A_a|_{D_W^{\text{cl}}}$ form an invariant $*$ -representation. In particular,

$$\pi_W^{\text{cl}}(a)\pi_W^{\text{cl}}(b)\xi = \pi_W^{\text{cl}}(ab)\xi, \quad \langle \pi_W^{\text{cl}}(a)\xi, \eta \rangle = \langle \xi, \pi_W^{\text{cl}}(a^*)\eta \rangle. \quad (1.49)$$

Each action is continuous in the graph topology. For each a , the nonnegative form

$$q_a([b]) = W(b^* a^* ab) = \|\pi_W(a)[b]\|^2$$

is closable on H . Its closure is $\bar{q}_a(\xi) = \|A_a \xi\|^2$ on $\mathcal{D}(A_a)$. Thus (1.48) is exactly the intersection of these closed form domains. Each $\pi_W^{\text{cl}}(a)$ has closure A_a .

Proof. A graph Cauchy net (ξ_i) has a Hilbert limit ξ and limits $\eta_a = \lim_i \pi_W(a)\xi_i$. Closedness of A_a gives $\xi \in \mathcal{D}(A_a)$ and $A_a \xi = \eta_a$. If $\xi = 0$, closability gives $\eta_a = 0$ for every a . Hence the graph completion injects into the right side of (1.48).

To prove the reverse inclusion, fix ξ in that intersection and a finite set $F \subset \mathfrak{A}$. Set $c = \sum_{a \in F} a^* a$. For $u \in D$, the adjoint identity gives the estimate

$$\sum_{a \in F} \|\pi_W(a)u\|^2 = \langle \pi_W(c)u, u \rangle \leq \|\pi_W(c)u\| \|u\|. \quad (1.50)$$

Since $\xi \in \mathcal{D}(A_c)$, choose $u_j \in D$ with $u_j \rightarrow \xi$ and $\pi_W(c)u_j \rightarrow A_c \xi$. Apply (1.50) to $u_j - u_k$. Every $\pi_W(a)u_j$, $a \in F$, is Cauchy. Closedness identifies its limit with $A_a \xi$. Thus D approximates ξ in any prescribed finite collection of graph seminorms. Directing these approximations by finite sets and positive tolerances produces one net converging in all graph seminorms.

Let this net be (ξ_i) and fix $b \in \mathfrak{A}$. Then $\pi_W(b)\xi_i \rightarrow A_b \xi$ and $\pi_W(a)\pi_W(b)\xi_i = \pi_W(ab)\xi_i \rightarrow A_{ab}\xi$ for every a . Closedness gives $A_b \xi \in D_W^{\text{cl}}$ and $A_a A_b \xi = A_{ab}\xi$. Linearity follows by the same approximation. Approximating both vectors in the algebraic adjoint identity proves the second equality in (1.49). Continuity follows from $\|A_a A_b \xi\| = \|A_{ab}\xi\|$ and $\|A_b \xi\|$ itself.

The form norm for q_a is $(\|\xi\|^2 + \|\pi_W(a)\xi\|^2)^{1/2}$. Its completion is the graph of A_a , which proves the form assertion. Finally $\pi_W(a) \subseteq \pi_W^{\text{cl}}(a) \subseteq A_a$, so taking closures proves the last statement. Any closed $*$ -representation extending π_W contains the limits of these graph Cauchy nets and agrees with their multiplier limits. Thus it extends π_W^{cl} , which proves minimality. $\square$

The word “closed” refers to completeness in the common graph topology. It does not assert that each restriction to D_W^{cl} is a closed operator or is self-adjoint. Example I.12.1 separates the common domain from the maximal domain of one multiplier.

I.13 NORM AND ORDER IN A COMPLETE INVOLUTIVE ALGEBRA

Bounded target algebras impose estimates which an arbitrary algebraic representation need not satisfy. We derive those estimates from the norm before using them in transfer or unitization. A Banach space is a complete normed vector space. A C^* -algebra is a complex $*$ -algebra complete in a norm satisfying $\|ab\| \leq \|a\|\|b\|$ and $\|a^*a\| = \|a\|^2$. The involution is isometric: the latter identity first gives $\|a\| \leq \|a^*\|$, and application to a^* gives equality. No representation by bounded operators is assumed in this definition.

LEMMA I.13.1 (*Adjoining a unit with its norm*). For a nonunital C^* -algebra A , the algebra $A^+ = A \oplus \mathbb{C}1$ with

$$(a + \lambda 1)(b + \mu 1) = ab + \lambda b + \mu a + \lambda\mu 1, \quad (a + \lambda 1)^* = a^* + \bar{\lambda} 1$$

has a complete C^* -norm extending the norm of A . If $A \neq \{0\}$, that norm is

$$\|a + \lambda 1\|_+ = \sup_{\|b\| \leq 1} \|ab + \lambda b\|.$$

For $A = \{0\}$ take $A^+ = \mathbb{C}$ with the absolute value.

Proof. For $A = \{0\}$ the assertion is the usual C^* -norm on $\mathbb{C}$. Assume henceforth that $A \neq \{0\}$. Write $L_x b = xb$ for $x \in A^+$ and $b \in A$. These are bounded maps on the Banach space A . For $a \in A$, the choice $b = a^*/\|a\|$ shows $\|L_a\| = \|a\|$ when $a \neq 0$. If $L_{a+\lambda 1} = 0$ with $\lambda \neq 0$, then $-a/\lambda$ is a left identity. Its adjoint is a right identity; multiplying the two identities shows that they agree and give a unit in A , a contradiction. Thus $x \mapsto L_x$ is injective and the displayed expression is a norm.

For $b \in A$,

$$\|xb\|^2 = \|b^*x^*xb\| \leq \|b\|\|x^*xb\| \leq \|L_{x^*x}\|\|b\|^2.$$

Taking suprema gives $\|L_x\|^2 \leq \|L_{x^*x}\| \leq \|L_{x^*}\|\|L_x\|$. Applying this also to x^* proves equality of the two adjoint norms and then the C^* -identity.

To check completeness directly, one has $|\lambda| \leq \|L_{a+\lambda 1}\|$. Otherwise $L_a = L_{a+\lambda 1} - \lambda 1$ would be invertible by the convergent geometric series. Choose $e \in A$ with $ae = a$. Then $L_a L_e = L_a$ forces $L_e = I$, again giving a unit in A . It follows that

$$\|a + \lambda 1\|_+ \leq \|a\| + |\lambda| \leq 3\|a + \lambda 1\|_+.$$

A Cauchy sequence therefore has Cauchy components in A and $\mathbb{C}$, whose limits supply its limit in A^+ . $\square$

For an element h of a nonzero unital Banach algebra, its spectrum $\sigma(h)$ is the set of complex λ for which $\lambda 1 - h$ has no two-sided inverse. In a nonunital C^* -algebra we use the unitization just constructed; in particular, the spectrum for $A = \{0\}$ is computed in $A^+ = \mathbb{C}$. In every one of these nonzero ambient algebras, $\lambda 1$ is invertible exactly when $\lambda \neq 0$: its inverse is $\lambda^{-1} 1$, while zero cannot have an inverse because $1 \neq 0$. Thus $\sigma(0) = \{0\}$. A self-adjoint element is called positive when its spectrum is contained in $[0, \infty)$. The next two lemmas show that this is precisely the algebraic notion of a square c^*c .

LEMMA 1.13.2 (*The calculus of a self-adjoint element*). For a self-adjoint h in a C^* -algebra, the spectrum is real and contained in $[-\|h\|, \|h\|]$. Every continuous complex function f on this interval has a well-defined value $f(h)$ obtained by norm limits of polynomials. The construction preserves sums, products, constants, and conjugation, and

$$\|f(h)\| = \max_{t \in \sigma(h)} |f(t)|.$$

If A is nonunital, $h \in A$, and $f(0) = 0$, then $f(h) \in A$. A continuous function specified on a smaller interval containing the spectrum may be extended constantly past its endpoints. Its value is independent of that extension by the norm identity. Every positive element has a positive square root in the same algebra. Every self-adjoint element is a difference $h = h_+ - h_-$ of positive elements with $h_+ h_- = 0$ and $\|h_{\pm}\| \leq \|h\|$.

Proof. For $h = 0$, take $f(h) = f(0)1$. We may therefore assume $\|h\| > 0$. We supply the spectral facts used in the polynomial construction. In a nonzero unital Banach algebra, the series $(\lambda 1 - x)^{-1} = \sum_{n \geq 0} \lambda^{-n-1} x^n$ converges for $|\lambda| > \|x\|$; the same series about any inverse proves openness of the invertible set and holomorphy of the resolvent. Thus the spectrum is compact. It is nonempty: an everywhere-defined resolvent would be an entire bounded Banach-valued function tending to zero at infinity. The circle derivative estimate makes such a function constant, hence zero, contradicting its inverse identity. The circle formulas of Lemma 1.2.2 hold for Banach-valued functions by the same uniform Riemann sums and norm estimates; completeness supplies their limits.

Polynomial factorization shows $\sigma(p(x)) = p(\sigma(x))$. Indeed, the factors of $p(x) - \lambda 1$ commute, and a product of commuting elements is invertible exactly when all its factors are invertible. Existence of complex roots for a nonconstant polynomial follows from the same circle estimate: otherwise its reciprocal would be bounded entire and constant. Repeated division gives the factorization used here. If $r(x) = \max\{|\lambda| : \lambda \in \sigma(x)\}$ and $R > r(x)$, deformation of a circle from radius greater than $\|x\|$ to radius R gives

$$x^n = \frac{1}{2\pi i} \int_{|\lambda|=R} \lambda^n (\lambda 1 - x)^{-1} d\lambda.$$

The norm estimate and the polynomial spectral identity imply $r(x) = \lim_n \|x^n\|^{1/n}$. For a normal element x , meaning $x^*x = xx^*$, the C^* -identity gives $\|x^2\| = \|x\|^2$. Iteration at powers 2^n therefore gives $r(x) = \|x\|$.

For self-adjoint h , the exponential series gives $(e^{ith})^* e^{ith} = 1$, so $\|e^{ith}\| = 1$ for real t . When $\text{Im } \lambda > 0$, the norm-convergent integral

$$i \int_0^\infty e^{-it(h-\lambda 1)} dt$$

is an inverse of $h - \lambda 1$, by integration of its derivative. Conjugation gives an inverse in the lower half-plane. Hence the spectrum of h is real. For any polynomial p , the element $p(h)$ is normal; thus $\|p(h)\| = \max_{\sigma(h)} |p|$.

Continuous functions on a compact real interval admit uniform polynomial approximation. An explicit proof on $[0, 1]$ uses $B_n f(t) = \sum_{k=0}^n f(k/n) \binom{n}{k} t^k (1-t)^{n-k}$. The binomial coefficients with these weights have total mass one, mean t , and variance $t(1-t)/n$. Splitting the sum into $|k/n - t| < \delta$ and its complement bounds the approximation error by

$$\sup_{|u-v|<\delta} |f(u) - f(v)| + \frac{2\|f\|_\infty}{4n\delta^2}.$$

Uniform continuity, followed by $n \rightarrow \infty$, proves the assertion; an affine change of variable treats the stated interval. The polynomial norm identity now defines $f(h)$ by completion and proves all its displayed properties. Subtracting the constant term from the approximating polynomials proves the assertion when $f(0) = 0$.

If h is positive, take $f(t) = \sqrt{\max(t, 0)}$ on the full interval. Then $f(h)^2 = h$, since the difference vanishes on $\sigma(h)$. The element $f(h)$ is itself positive: write it as $q(h)^2$ with $q(t) = (\max(t, 0))^{1/4}$, and use the real spectrum of the self-adjoint $q(h)$ and the polynomial spectral identity. Taking $f_\pm(t) = \max(\pm t, 0)$ gives the final decomposition and norm bounds by the same argument. $\square$

LEMMA 1.13.3 (*Positive squares and order bounds*). The positive elements form a norm-closed cone whose intersection with its negative is zero. They are exactly the elements c^*c . Write $a \leq b$ when $b - a$ is positive. Conjugation by any element preserves this order, and

$$0 \leq a \leq b \implies \|a\| \leq \|b\|, \quad c^*c \leq \|c\|^2 1.$$

Proof. For self-adjoint h and $R > 0$, the condition $\|h - R1\| \leq R$ says exactly $\sigma(h) \subset [0, 2R]$. This criterion and the triangle inequality show that positive elements are closed under addition: use radii $\|a\|$ and $\|b\|$ for two nonzero positive elements. Scalar multiplication by nonnegative reals preserves positivity. A norm-convergent sequence of positive elements has a common bounding radius, so the cone is closed. If both h and $-h$ are positive, the spectrum is $\{0\}$ and the normal-element norm identity gives $h = 0$.

For every x , the nonzero spectra of x^*x and xx^* coincide. This follows by scaling the identity $(1 - uv)^{-1} = 1 + u(1 - vu)^{-1}v$ whenever the latter inverse exists. Suppose first that $-x^*x$ is positive. Writing $x = u + iv$ with self-adjoint u, v gives

$$xx^* = 2u^2 + 2v^2 - x^*x \geq 0.$$

The common nonzero spectrum must then be both positive and negative, hence empty. Thus $\|x\|^2 = \|x^*x\| = 0$. For a general x , put $h = x^*x$, take its negative part h_- from Lemma 1.13.2, and set $y = xh_-^{1/2}$. Commutation in the calculus gives $y^*y = hh_- = -h_-$. The preceding argument forces $y = 0$, hence $h_- = 0$ and $x^*x \geq 0$. Conversely, a positive element is the square of its self-adjoint square root, proving the characterization.

If $h = d^*d \geq 0$, then $c^*hc = (dc)^*(dc) \geq 0$. The spectral bound for c^*c is $[0, \|c\|^2]$, which proves the second order estimate. Finally $b \leq \|b\|1$ implies $a \leq \|b\|1$ when $0 \leq a \leq b$. The spectra of a and $\|b\|1 - a$ are nonnegative, so $\sigma(a) \subset [0, \|b\|]$ and $\|a\| \leq \|b\|$. $\square$

LEMMA 1.13.4 (*A positive approximate identity*). Every C^* -algebra $\mathcal{A}$ has a net (e_i) of positive elements with $\|e_i\| \leq 1$ and $e_i a \rightarrow a, a e_i \rightarrow a$ for every $a \in \mathcal{A}$. Such a net is called a positive contractive two-sided approximate identity.

Proof. For a unital algebra take the constant net 1; the zero algebra is immediate. Suppose now that $\mathcal{A}$ is nonunital. For a finite set $F \subset \mathcal{A}$ and $\epsilon > 0$, define in the unitization

$$t_F = \sum_{a \in F} (a^* a + a a^*), \quad e_{F,\epsilon} = t_F(t_F + \epsilon 1)^{-1}.$$

The calculus shows $e_{F,\epsilon} \in \mathcal{A}$ and $0 \leq e_{F,\epsilon} \leq 1$, since $t/(t + \epsilon)$ lies in $[0, 1]$ for $t \geq 0$ and vanishes at zero. For $a \in F$, both $a^* a$ and $a a^*$ are bounded above by t_F . Writing $e = e_{F,\epsilon}$, Lemma 1.13.3 gives

$$\|a(1 - e)\|^2 \leq \|(1 - e)t_F(1 - e)\| \leq \sup_{t \geq 0} \frac{\epsilon^2 t}{(t + \epsilon)^2} = \frac{\epsilon}{4}.$$

Apply the same bound to a^* to obtain $\|(1 - e)a\|^2 \leq \epsilon/4$. Direct finite sets by inclusion and positive ϵ by decreasing size. Every fixed a then satisfies both limits. $\square$

1.14 DAGGER TRANSFER AND COMPLETION

THEOREM 1.14.1 (*Dagger transfer*). Let $T : \mathfrak{A} \rightarrow \mathfrak{B}$ be a $*$ -homomorphism of complex $*$ -algebras, let $\omega : \mathfrak{B} \rightarrow \mathbb{C}$ be positive, and put $\mathcal{W} = \omega \circ T$. Then

$$\langle a, c \rangle_{\mathcal{W}} = \langle T(a), T(c) \rangle_{\omega}, \quad \mathcal{N}_{\mathcal{W}} = T^{-1}(\mathcal{N}_{\omega}). \quad (1.51)$$

Consequently

$$U_0 : D_{\mathcal{W}} \longrightarrow D_{\omega}, \quad [a] \longmapsto [T(a)], \quad (1.52)$$

is an isometric injection and extends uniquely to an isometry

$$U : \mathcal{H}_{\mathcal{W}} \longrightarrow \mathcal{H}_{\omega} \quad (1.53)$$

with closed range. On the algebraic common domain,

$$U_0 \pi_{\mathcal{W}}(a) = \pi_{\omega}(T(a)) U_0. \quad (1.54)$$

If $\mathfrak{B}$ is a C^* -algebra, both sides of (1.54) extend boundedly and

$$\|\pi_{\omega}(b)\| \leq \|b\|, \quad \|\pi_{\mathcal{W}}(a)\| \leq \|T(a)\|, \quad (1.55)$$

while

$$U \pi_{\mathcal{W}}(a) = \pi_{\omega}(T(a)) U \quad \text{on } \mathcal{H}_{\mathcal{W}}. \quad (1.56)$$

Proof. For $a, c \in \mathfrak{A}$,

$$\mathcal{W}(c^* a) = \omega(T(c)^* T(a)),$$

which proves positivity, (1.51), and the isometry (1.52). Every isometry of pre-Hilbert spaces extends uniquely to their completions; its range is closed because its domain is complete. Multiplicativity of T gives (1.54).

Suppose now that $\mathfrak{B}$ is a C^* -algebra. Lemma 1.13.3, with conjugation by d , gives

$$\omega(d^* b^* b d) \leq \|b\|^2 \omega(d^* d),$$

which proves the first estimate in (1.55). Taking $b = T(a)$ and $d = T(c)$ proves the second. Continuity then extends (1.54) to (1.56). No equality between arbitrary unbounded closure domains is used. $\square$

PROPOSITION 1.14.2 (*Transfer of closed graph domains*). Use the data of Theorem 1.14.1, and set $K = U\mathcal{H}_W \subset \mathcal{H}_\omega$ and $E = U_0 D_W$. On the dense subspace $E \subset K$, define

$$\rho(a) = \pi_\omega(T(a))|_E.$$

Then ρ is a $*$ -representation, and

$$UD_W^{\text{cl}} = D_\rho^{\text{cl}}, \quad U\pi_W^{\text{cl}}(a)\xi = \rho^{\text{cl}}(a)U\xi. \quad (1.57)$$

Here every closure defining D_ρ^{cl} is taken in K . For the ambient closures on $\mathcal{H}_\omega$ one has

$$U\mathcal{D}(\overline{\pi_W(a)}) \subseteq \mathcal{D}(\overline{\pi_\omega(T(a))}), \quad \overline{\pi_\omega(T(a))}U\xi = U\overline{\pi_W(a)}\xi. \quad (1.58)$$

If T is surjective, U is unitary and $UD_W^{\text{cl}} = D_\omega^{\text{cl}}$. Without surjectivity this last domain equality need not hold, even when U is unitary.

Proof. The algebraic intertwining identifies $\rho(a)$ with $U\pi_W(a)U^{-1}$ on E . As a map onto K , U is unitary. It identifies the individual graph closures and all graph seminorms. Theorem 1.12.2 therefore gives (1.57). If $\xi_j \in D_W$ approaches ξ in the graph norm of $\pi_W(a)$, its images approach $U\xi$ and have multiplier images approaching $U\overline{\pi_W(a)}\xi$. This proves (1.58). For surjective T , one has $U_0 D_W = D_\omega$ and $T(\mathfrak{A}) = \mathfrak{B}$. Hence the Hilbert spaces and their full graph topologies agree. $\square$

Example 1.14.3 (*A Hilbert isometry can miss the full target domain*). Let $\mathfrak{A} = c_{00}(\mathbb{N})$ and $\mathfrak{B} = \mathbb{C}[n] + c_{00}(\mathbb{N})$, with pointwise multiplication and conjugation. Here $\mathbb{C}[n]$ denotes polynomial sequences on the positive integers. Let T be inclusion and set

$$\omega(b) = \sum_{n \geq 1} e^{-n} b_n, \quad W = \omega|_{\mathfrak{A}}.$$

All sums converge, and both functionals are positive and faithful: their square values are zero only on the zero element, since every weight is positive. Both GNS completions identify with $H = \ell^2(\mathbb{N}, e^{-n})$, because they contain the finite sequences. Thus U is the identity on H . Every $\mathfrak{A}$ -multiplier is bounded, so $D_W^{\text{cl}} = H$. The element $n \in \mathfrak{B}$ acts by an unbounded multiplier. Finite truncations give cores for every polynomial multiplier. Therefore

$$D_\omega^{\text{cl}} = \left\{ v : \sum_{n \geq 1} e^{-n} n^{2k} |v_n|^2 < \infty \text{ for all } k \geq 0 \right\}.$$

The vector $v_n = e^{n/2}/n$ belongs to H , but fails the condition for $k = 1$. Thus $UD_W^{\text{cl}} \not\subseteq D_\omega^{\text{cl}}$. Only the restricted algebra $T(\mathfrak{A})$ appears in (1.57).

For a field interpretation, algebra products represent repeated operations on a common domain. Theorem 1.12.2 supplies this domain after positivity has supplied the Hilbert norm. Proposition 1.14.2 preserves it for the represented algebra. A vacuum vector, self-adjoint generators, locality, and a positive energy condition require their own constructions.

1.15 POSITIVE UNITIZATION FOR NONUNITAL C^* -ALGEBRAS

The unit missing from a convolution algebra cannot be adjoined to an arbitrary positive functional without an additional bound. Let ω be a positive linear functional on a nonunital $*$ -algebra $\mathfrak{A}$. Write $\mathfrak{A}^+ = \mathfrak{A} \oplus \mathbb{C}1$ and

$$\omega_m(a + \lambda 1) = \omega(a) + m\lambda. \quad (1.59)$$

Expanding a square in this algebra gives

$$\omega_m((a + \lambda 1)^*(a + \lambda 1)) = \omega(a^*a) + \lambda\omega(a^*) + \bar{\lambda}\omega(a) + m|\lambda|^2.$$

Requiring real nonnegative values first forces $m \geq 0$ and Hermitian symmetry. For $m > 0$, minimization in λ gives the inequality below. For $m = 0$, varying λ forces $\omega(a) = 0$ for every a . Thus the exact algebraic criterion is

$$\omega_m \text{ is positive} \iff \begin{cases} m \in \mathbb{R}, & m \geq 0, \\ \omega(a^*) = \overline{\omega(a)}, \\ |\omega(a)|^2 \leq m \omega(a^*a) & (a \in \mathfrak{A}). \end{cases} \quad (1.60)$$

A bounded representation is nondegenerate when $\overline{\text{span}\{\pi(a)x : a \in \mathfrak{A}, x \in H\}} = H$. It is cyclic with vector Ω when $\overline{\pi(\mathfrak{A})\Omega} = H$.

THEOREM 1.15.1 (*The nonunital C^* case*). Let $\mathfrak{A}$ be a nonunital C^* -algebra and let $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ be positive. Then ω is Hermitian and bounded, and the functional (1.59) on the unitization constructed in Lemma 1.13.1 is positive exactly when

$$m \in \mathbb{R}, \quad m \geq \|\omega\|. \quad (1.61)$$

Let (e_i) be any positive contractive two-sided approximate identity. For the minimal extension $m = \|\omega\|$, its GNS vector $\Omega = [1]$ satisfies

$$[e_i] \longrightarrow \Omega, \quad (1.62)$$

and the restricted representation of $\mathfrak{A}$ is bounded, nondegenerate, and cyclic:

$$\omega(a) = \langle \pi_\omega(a)\Omega, \Omega \rangle, \quad \overline{\pi_\omega(\mathfrak{A})\Omega} = \mathcal{H}_\omega. \quad (1.63)$$

For $m > \|\omega\|$, the vector $[1]$ of that larger extension is not the limit in (1.62).

Proof. Every self-adjoint element is the difference of two positive elements, so positivity makes ω Hermitian. If ω were unbounded, its values would be unbounded on the positive unit ball: otherwise the decomposition of an arbitrary element into four positive parts would bound it. One could then choose $a_n \geq 0$ with

$$\|a_n\| \leq 2^{-n}, \quad \omega(a_n) = 1.$$

The norm-convergent positive sum $a = \sum_n a_n$ would satisfy $\omega(a) \geq \sum_{n=1}^N \omega(a_n) = N$ for every N , a contradiction. Hence ω is bounded.

Lemma 1.13.4 constructs a positive contractive approximate identity. Cauchy–Schwarz and $e_i a \rightarrow a$ give

$$|\omega(a)|^2 = \lim_i |\omega(e_i a)|^2 \leq \|\omega\| \omega(a^*a). \quad (1.64)$$

Equations (1.60) and (1.64) prove positivity for $m = \|\omega\|$; adding $(m - \|\omega\|)$ times the positive quotient functional $a + \lambda \mathbf{1} \mapsto \lambda$, whose value on a square is $|\lambda|^2$, proves sufficiency for every larger m . For a positive functional η on a unital C^* -algebra, Cauchy–Schwarz and Lemma 1.13.3 give

$$|\eta(a)|^2 \leq \eta(\mathbf{1})\eta(a^*a) \leq \eta(\mathbf{1})^2\|a\|^2.$$

Evaluation at $\mathbf{1}$ therefore gives $\|\eta\| = \eta(\mathbf{1})$. Applying this to $\eta = \omega_m$ and restricting yields $\|\omega\| \leq m$.

It remains to identify the vector $[\mathbf{1}]$. If $\|\omega\| > 0$, choose $\|a\| \leq 1$ with $|\omega(a)|$ arbitrarily close to $\|\omega\|$. Cauchy–Schwarz gives $|\omega(e_i a)|^2 \leq \omega(e_i^2)\omega(a^*a) \leq \omega(e_i^2)\|\omega\|$. Since $e_i a \rightarrow a$, this forces the lower limit of $\omega(e_i^2)$ to be at least $\|\omega\|$. The upper bound is $\|\omega\|\|e_i^2\| \leq \|\omega\|$. Therefore $\omega(e_i^2) \rightarrow \|\omega\|$. Since $0 \leq e_i^2 \leq e_i \leq \mathbf{1}$ in the unitization, also $\omega(e_i) \rightarrow \|\omega\|$. Therefore

$$\|[\mathbf{1}] - [e_i]\|^2 = \|\omega\| - 2\omega(e_i) + \omega(e_i^2) \rightarrow 0,$$

which is (1.62); the zero functional is immediate. Moreover $\pi_\omega(a)\Omega = [a]$, proving cyclicity, and $\pi_\omega(e_i)[a] = [e_i a] \rightarrow [a]$, proving nondegeneracy. The C^* -estimate in (1.55) proves boundedness. If $m > \|\omega\|$, the same computation instead gives

$$\lim_i \|[\mathbf{1}] - [e_i]\|^2 = m - \|\omega\| > 0. \quad \square$$

For $\mathfrak{A} = \{0\}$, the only functional is $\omega = 0$, and $\mathfrak{A}^+ = \mathbb{C}$. The extension $\omega_m(\lambda) = m\lambda$ is positive exactly when $m \geq 0$. At minimal mass $m = 0$, its GNS Hilbert space and Ω are both zero, and the constant approximate identity $e_i = 0$ gives $[e_i] = \Omega$. For $m > 0$, the extended GNS space is one-dimensional with $\|\Omega\|^2 = m$; the original zero algebra acts as zero and $[e_i] = 0$ cannot converge to Ω .

Example 1.15.2 (Positive unitization can fail algebraically). Let $\mathfrak{A} = c_{00}(\mathbb{N})$ with pointwise multiplication and conjugation, and define

$$\omega(a) = \sum_{n \geq 1} n a_n.$$

This is a Hermitian positive functional. If a positive extension ω_m existed, Cauchy–Schwarz in the unitization, applied to the n th point mass e_n , would give

$$n^2 = |\omega_m(e_n)|^2 \leq \omega_m(\mathbf{1})\omega_m(e_n^*e_n) = mn$$

for every n , which is impossible for finite m .

LEMMA 1.15.3 (*The logarithmic size of the gamma term*). For each fixed $\sigma > 0$,

$$\operatorname{Re} \psi(\sigma + it) = \log |t| + O_\sigma(1) \quad (|t| \geq 1).$$

Consequently $k_\infty(t) = \operatorname{Re} \psi(1/4 + it/2) - \log \pi$ differs from $\log |t|$ by a bounded function outside $[-1, 1]$.

Proof. For $|t| \geq 1$, put $h_t(x) = (x + \sigma)/((x + \sigma)^2 + t^2)$ on $x \geq 0$. Its derivative changes sign at most once, and its maximum is at most $1/(2|t|)$. Thus $\int_0^\infty |h'_t(x)| dx \leq 1/|t|$. On each unit interval the fundamental theorem of calculus gives

$$\left| h_t(n) - \int_n^{n+1} h_t(x) dx \right| \leq \int_n^{n+1} |h'_t(x)| dx.$$

The real part of the series in (1.12) therefore equals

$$-\gamma + \lim_{N \rightarrow \infty} \left(\sum_{n=0}^N \frac{1}{n+1} - \frac{1}{2} \log \frac{(N+1+\sigma)^2 + t^2}{\sigma^2 + t^2} \right) + O(|t|^{-1}).$$

By the defining limit of γ , this is $\frac{1}{2} \log(\sigma^2 + t^2) + O(|t|^{-1})$. The stated bound follows, and replacing t by $t/2$ changes the logarithm only by $-\log 2$. $\square$

1.16 THE GAUSSIAN OBSTRUCTION TO FINITE MASS

For the full Weil functional, failure of positive unitization follows from its arithmetic local terms. It does not require a hypothesis about the zeros. The relevant functions approach the convolution identity while remaining inside $\mathcal{A}_\zeta$:

$$g_a(u) = \frac{e^{-u^2/(4a)}}{\sqrt{4\pi a}}, \quad \widehat{g}_a(t) = e^{-at^2}, \quad g_a^* \star g_a = g_{2a}, \quad a > 0. \quad (1.65)$$

THEOREM 1.16.1 (*No finite positive unitization of the Weil functional*). As $a \downarrow 0$, the functional in (1.39) satisfies

$$L_\zeta(g_a) = \frac{a^{-1/2} \log(1/a)}{4\sqrt{\pi}} + O(a^{-1/2}). \quad (1.66)$$

In particular $L_\zeta(g_{2a}) > 0$ for all sufficiently small a , and

$$\frac{|L_\zeta(g_a)|^2}{L_\zeta(g_a^* \star g_a)} \sim \frac{\sqrt{2}}{4\sqrt{\pi}} a^{-1/2} \log(1/a) \longrightarrow \infty. \quad (1.67)$$

There is no positive extension of L_ζ to $\mathcal{A}_\zeta^+$ with a finite value at the adjoined unit.

Proof. Every derivative of g_a is a polynomial times a Gaussian. Thus $g_a \in \mathcal{A}_\zeta$. Completing the square proves (1.65), with the Fourier normalization (1.2). The polar contribution is $2e^{a/4} = O(1)$.

For $0 < a \leq 1$, split the Gaussian exponent into two equal parts. Since $\Lambda_{\text{vM}}(n) \leq \log n$, the absolute prime contribution is bounded by

$$\frac{e^{-(\log 2)^2/(8a)}}{\sqrt{\pi a}} \sum_{n \geq 2} \frac{\log n}{\sqrt{n}} e^{-(\log n)^2/8}. \quad (1.68)$$

The series is finite: its terms are at most n^{-2} for all sufficiently large n . Hence this contribution is exponentially small relative to $a^{-1/2}$.

Put $k_\infty(t) = \operatorname{Re} \psi(1/4 + it/2) - \log \pi$. Lemma 1.15.3 shows that $k_\infty(t) - \log |t|$ is bounded for $|t| \geq 1$. It is integrable on $[-1, 1]$, since k_∞ is smooth there. Therefore

$$\int_{\mathbb{R}} e^{-at^2} (k_\infty(t) - \log |t|) dt = O(a^{-1/2}).$$

Substitution $s = \sqrt{a} t$ gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-at^2} \log |t| dt = \frac{a^{-1/2}}{2\pi} \left(\frac{\sqrt{\pi}}{2} \log(1/a) + \int_{\mathbb{R}} e^{-s^2} \log |s| ds \right).$$

The remaining integral is finite. Adding all three local terms proves (1.66). Applying that formula at a and $2a$ proves (1.67). A positive unitization of mass m would require $|L_\zeta(g_a)|^2 \leq m L_\zeta(g_{2a})$ by (1.60), which contradicts the divergence. $\square$

THEOREM 1.16.2 (*Positive finite Gaussian subspaces*). Fix distinct positive real numbers $\lambda_1, \dots, \lambda_r$. There exists $a_0 > 0$ such that the Weil form is positive definite on

$$E_a = \text{span}_{\mathbb{C}}\{g_{a\lambda_1}, \dots, g_{a\lambda_r}\} \quad (0 < a < a_0).$$

The dimension of E_a is r . This assertion is unconditional. The threshold a_0 may depend on the entire finite family.

Proof. The Gram matrix of the displayed functions is the real symmetric matrix

$$G(a)_{jk} = L_{\mathcal{G}}(g_{a(\lambda_j + \lambda_k)}).$$

Apply (1.66) at each of the finitely many scales $a(\lambda_j + \lambda_k)$. Entry by entry,

$$\frac{4\sqrt{\pi}\sqrt{a}}{\log(1/a)} G(a) \longrightarrow M, \quad M_{jk} = \frac{1}{\sqrt{\lambda_j + \lambda_k}}. \quad (1.69)$$

For a column $c \in \mathbb{C}^r$, let c^* be its conjugate transpose. Then

$$c^* M c = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \left| \sum_{j=1}^r c_j e^{-\lambda_j t^2} \right|^2 dt. \quad (1.70)$$

If this integral vanishes, continuity makes its integrand zero everywhere. Order the λ_j increasingly, divide the resulting identity by $e^{-\lambda_1 t^2}$, and let $t \rightarrow +\infty$. This gives $c_1 = 0$. Repeating proves $c = 0$. Thus M is positive definite. Write $\|c\|_2^2 = \sum_j |c_j|^2$ and put

$$\mu = \inf_{\|c\|_2=1} c^* M c.$$

This infimum is attained: from a minimizing sequence, successively choose subsequences on which each of its finitely many real and imaginary coordinates converges. A bounded real sequence has such a subsequence by bisecting a bounded interval and retaining at each step a half containing infinitely many terms. The nested intervals have lengths tending to zero, and completeness supplies their common limit. The resulting coordinate limit still has norm one, and continuity of the finite quadratic expression gives its value μ . Positive definiteness at that nonzero limit therefore gives $\mu > 0$.

Let $A(a)$ denote the matrix on the left of (1.69), and set $\delta(a) = \max_{j,k} |A(a)_{jk} - M_{jk}|$. The entrywise convergence gives $\delta(a) \rightarrow 0$. Finite-sum Cauchy–Schwarz gives, for every $c \in \mathbb{C}^r$,

$$|c^*(A(a) - M)c| \leq \delta(a) \left(\sum_j |c_j| \right)^2 \leq r \delta(a) \|c\|_2^2.$$

Choose $0 < a < 1$ small enough that $r \delta(a) < \mu/2$. Then $c^* A(a) c \geq (\mu/2) \|c\|_2^2$. The scalar relating $A(a)$ to $G(a)$ is positive, so $G(a)$ is positive definite as well. The same linear-independence argument applied to the Fourier transforms $e^{-a\lambda_j t^2}$ proves $\dim E_a = r$. $\square$

For $\lambda_1 = 1$ and $\lambda_2 = 4$, the limiting matrix is

$$M = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{5} \\ 1/\sqrt{5} & 1/\sqrt{8} \end{pmatrix}, \quad \det M = \frac{1}{20} > 0.$$

This gives a positive two-dimensional family directly from the local terms. The quantifiers in Theorem 1.16.2 fix the finite family before choosing its scale. They do not imply positivity on all of $\mathcal{A}_{\mathcal{G}}$ at once.

COROLLARY 1.16.3 (*Excluded positive carriers*). There is no representation

$$L_\zeta(g) = \langle \pi(g)\Omega, \Omega \rangle$$

on a positive Hilbert space with Ω of finite norm in a common invariant $*$ -representation domain. There is also no $*$ -homomorphism $T : \mathcal{A}_\zeta \rightarrow \mathfrak{B}$ to a C^* -algebra with a positive linear functional $\omega : \mathfrak{B} \rightarrow \mathbb{C}$ such that $L_\zeta = \omega \circ T$.

Proof. In the vector case, the adjoint identity on the invariant domain gives

$$|L_\zeta(g)|^2 \leq \|\Omega\|^2 \|\pi(g)\Omega\|^2 = \|\Omega\|^2 L_\zeta(g^* \star g).$$

This contradicts (1.67). In the C^* case, positivity implies boundedness and $|\omega(b)|^2 \leq \|\omega\| \omega(b^*b)$ by Theorem 1.15.1, or its unital Cauchy–Schwarz proof. Substitute $b = T(g)$ and use the same divergence. $\square$

This obstruction concerns the functional's representing vector. It does not prevent bounded GNS multipliers. Indeed, under RH, (1.43) and $|\widehat{f}(t)| \leq \|f\|_{L^1}$ give

$$L_\zeta((f \star b)^* \star (f \star b)) \leq \|f\|_{L^1}^2 L_\zeta(b^* \star b).$$

Thus every convolution multiplier is bounded under that hypothesis, and $D_{L_\zeta}^{\text{cl}} = \mathcal{H}_{L_\zeta}$. The functional itself has no finite representing vector. A positive functional on a proper test algebra, or a positive weight finite on a specified test ideal, remains compatible with this restriction. A positive weight assigns values in $[0, \infty]$ to positive elements, additively and homogeneously for nonnegative scalars. Its finite test ideal and all products must be specified before a GNS or field interpretation can be asserted.

1.17 WHY AN INDEFINITE QUOTIENT DOES NOT GIVE POSITIVITY

The Hermitian part of the construction exists without positivity. For a complex $*$ -algebra $\mathfrak{A}$ and a Hermitian functional W , meaning $W(a^*) = \overline{W(a)}$, put

$$B_W(f, g) = W(g^*f), \quad R_W = \{f : B_W(f, g) = 0 \text{ for every } g \in \mathfrak{A}\}.$$

Then R_W is a left ideal, since

$$B_W(af, g) = W(g^*af) = B_W(f, a^*g) = 0 \quad (f \in R_W).$$

The quotient $\mathfrak{A}/R_W$ consequently has a nondegenerate Hermitian pairing and an algebraic left action satisfying $B_W(af, g) = B_W(f, a^*g)$. This applies unconditionally to $(\mathcal{A}_\zeta, L_\zeta)$ by (1.41). It supplies neither a positive norm nor a class of the missing unit. Under positivity, Cauchy–Schwarz identifies R_W with $\mathcal{N}_W$, and Theorem 1.11.2 applies.

Let $V = \mathbb{C}^2$, $J = \text{diag}(1, -1)$, and

$$[x, y] = y^* J x.$$

The vectors $e_1 + e_2$ and $e_1 - e_2$ are isotropic, while their sum is not. Thus the isotropic cone $\{x : [x, x] = 0\}$ is not a linear subspace and cannot be divided out. The radical $\{x : [x, y] = 0 \forall y\}$ is zero here. In general, quotienting by the radical produces only a nondegenerate Hermitian space; it supplies neither a fundamental symmetry nor a complete Hilbert majorant. A fundamental symmetry is a linear involution

J satisfying $[Jx, y] = [x, Jy]$ and $[Jx, x] > 0$ for $x \neq 0$. A complete Hilbert majorant is a complete Hilbert norm on the same space for which the original Hermitian pairing is continuous. In this finite-dimensional example the displayed J gives both choices, but the radical quotient itself selects neither choice.

Nor is a vector functional for an indefinite adjoint positive. On $M_2(\mathbb{C})$ put $A^\sharp = JA^*J$, let $\Omega = e_1$, and set $\omega(A) = [A\Omega, \Omega]$. For

$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

one has

$$\omega(X^\sharp X) = [X\Omega, X\Omega] = [e_2, e_2] = -1. \quad (1.71)$$

Hence an indefinite pairing does not provide the positivity required in (1.42).

1.18 THE INDEPENDENT ARITHMETIC QUESTION

PROPOSITION 1.18.1 (*Formal factorization*). For a linear functional W on a complex $*$ -algebra $\mathfrak{A}$, the following are equivalent:

- (i) W is positive;
- (ii) there are a complex $*$ -algebra $\mathfrak{B}$ and maps

$$\omega : \mathfrak{B} \longrightarrow \mathbb{C}, \quad T : \mathfrak{A} \longrightarrow \mathfrak{B},$$

where ω is positive, T is a $*$ -homomorphism, and $W = \omega \circ T$.

Proof. Dagger transfer proves (ii) $\Rightarrow$ (i). For the converse take $\mathfrak{B} = \mathfrak{A}$, $\omega = W$, and T equal to the identity. $\square$

For $W = L_\zeta$, formal factorization is therefore equivalent to the positivity in (1.42), and hence to RH. The identity factorization, or a Hilbert space whose spectral data are prescribed from the zeta zeros, cannot establish that positivity independently.

Open Problem 1.18.2 (*Independent dagger realization of L_ζ*). Construct a complex $*$ -algebra $\mathfrak{B}$, a positive linear functional $\omega : \mathfrak{B} \rightarrow \mathbb{C}$, and a $*$ -homomorphism $T : \mathcal{A}_\zeta \rightarrow \mathfrak{B}$ such that

$$L_\zeta = \omega \circ T. \quad (1.72)$$

Derive the carrier, the map, and the positivity of ω from the polar, archimedean, and prime-power structures in (1.39), without using RH, the multiset of zeta zeros, or an assertion equivalent to (1.42). Prove (1.72) with all three local terms and the Fourier and measure conventions above. Corollary 1.16.3 excludes a positive functional on a C^* -algebra as the target. Any admissible positive functional ω must fail the finite-mass unitization bound on the image of T . Theorem 1.12.2 and Proposition 1.14.2 then specify the common invariant domain supplied by this factorization.

The resulting representation would still need an additional comparison to a spectral determinant. Equality of functionals does not preserve trace multiplicities. For example, on $\mathbb{C}$ the positive functional $W(z) = mz$, with integer $m \geq 2$, has a one-dimensional GNS space. A scalar λ consequently has determinant $s - \lambda$ there. The trace realization $z \mapsto zI_m$ on $\mathbb{C}^m$ has the same functional and determinant $(s - \lambda)^m$. Thus a determinant realization must identify the operator, its domain, and spectral multiplicities separately.

1.19 FINITE MATRIX KERNELS AND POSITIVE TRACE

A scalar obtained by closing an operator kernel need not define a positive pairing. This distinction can already be seen on finite matrices inside the formal operator algebra. For this comparison, use the underlying field $\mathbb{C}$ with the discrete topology and put

$$H_{\text{for}} = \mathbb{C}[[x, y]]/\mathbb{C}.$$

Enumerate the nonconstant monomials as $e_o, e_1, \dots$ in increasing total degree. The formal topology makes H_{for} the product $\mathbb{C}^{\mathbb{N}}$ of discrete coefficient spaces. Write $\epsilon^j(h) = h_j$ for its coefficient functionals. A continuous endomorphism has a matrix with finitely many nonzero entries in each row. Indeed, every output coefficient must depend on finitely many input coefficients. Conversely, this condition makes every finite output projection continuous. Denote this algebra by $\mathcal{E}$. Its operator topology requires agreement of each entire row in a fixed finite set of output rows.

Put $E_{ij} = e_i \otimes \epsilon^j$ and let $\mathcal{F}_o$ be their finite complex span. These are the matrices with finitely many nonzero entries in total. They satisfy

$$E_{ij}E_{k\ell} = \delta_{jk}E_{i\ell}, \quad E_{ij}^* = E_{ji}.$$

Conjugating coefficients defines the displayed involution on $\mathcal{F}_o$. This algebra has no unit, since each of its elements annihilates all but finitely many basis vectors. It is dense in $\mathcal{E}$: retaining finitely many whole rows of a row-finite matrix gives a matrix in $\mathcal{F}_o$ with those rows unchanged.

The same matrix units act on $D_o = \mathbb{C}^{(\mathbb{N})}$, the finite sequences, inside the Hilbert space $\ell^2(\mathbb{N})$ over the usual complex field. Its inner product is $\langle v, w \rangle = \sum_j v_j \overline{w_j}$. Every finite matrix preserves D_o , and conjugate transpose is its Hilbert adjoint restricted to D_o . Thus this comparison preserves the algebra and involution on $\mathcal{F}_o$. It does not identify the two topologies or their completions.

PROPOSITION 1.19.1 (*The positive finite-matrix trace*). Define $\tau(B) = \sum_i B_{ii}$ on $\mathcal{F}_o$. It is a positive linear functional with zero GNS null space. Its GNS completion is $\ell^2(\mathbb{N} \times \mathbb{N})$, with $[B]$ identified with its coefficient array (B_{ij}) . The GNS action is left matrix multiplication, and

$$\|\pi_\tau(A)\| = \|A\|_{\ell^2 \rightarrow \ell^2} \quad (A \in \mathcal{F}_o).$$

Nevertheless, τ has no positive extension of finite mass to $\mathcal{F}_o^+ = \mathbb{C}I + \mathcal{F}_o$.

Proof. All sums in the algebraic calculation are finite, and

$$\tau(C^*B) = \sum_{i,j} B_{ij} \overline{C_{ij}}, \quad \tau(B^*B) = \sum_{i,j} |B_{ij}|^2.$$

The second expression vanishes exactly when $B = o$. For a norm-Cauchy sequence of finite arrays, every coefficient has a limit b_{ij} . A common norm bound bounds every finite sum of $|b_{ij}|^2$, so the limit array is square-summable. Pass to the coefficientwise limit in a finite sum of $|(B_n)_{ij} - (B_m)_{ij}|^2$, then take the supremum over the finite sums. The Cauchy estimate gives $\|B_n - b\| \rightarrow o$. Conversely, finite rectangular truncations of a square-summable array converge in this norm. This identifies the quotient and completion. Multiplication by A acts separately on each column, so

$$\|AB\|^2 = \sum_j \|AB_{\bullet j}\|_{\ell^2}^2 \leq \|A\|_{\ell^2 \rightarrow \ell^2}^2 \sum_j \|B_{\bullet j}\|_{\ell^2}^2.$$

The operator norm on the right is finite. The entrywise Cauchy–Schwarz inequality bounds it by $(\sum_{i,j} |A_{ij}|^2)^{1/2}$. Conversely, take a matrix B whose only nonzero column is a finite vector v . The ratio $\|AB\|/\|B\|$ is $\|Av\|/\|v\|$. Since A uses only finitely many input coordinates, such vectors compute its full operator norm. This proves equality.

For $n \geq 1$, let $Q_n = \sum_{i=0}^{n-1} E_{ii}$. It is a self-adjoint projection with $\tau(Q_n) = n$. If a positive extension had value c on I , then c would be real and nonnegative, while

$$0 \leq \tau^+((I - Q_n)^*(I - Q_n)) = c - n$$

for every n . No finite c satisfies these inequalities. Equivalently, $|\tau(Q_n)|^2/\tau(Q_n^*Q_n) = n$ is unbounded. $\square$

The map $(\lambda, B) \mapsto \lambda I + B$ identifies the algebraic unitization with the displayed subalgebra of $\mathcal{E}$. It is injective because a scalar multiple of the infinite identity is a finite matrix only when that scalar is zero. On this subalgebra, any linear extension of finite-matrix trace has the form

$$\tau_c(\lambda I + B) = c\lambda + \text{tr}(B).$$

For $A, B \in \mathcal{F}_0$, its value on a product is

$$\tau_c((\lambda I + A)(\mu I + B)) = c\lambda\mu + \lambda \text{tr}(B) + \mu \text{tr}(A) + \sum_{i,j} A_{ij}B_{ji}.$$

The sum is finite and is unchanged when the two factors are exchanged. Thus τ_c is cyclic, but the proposition excludes positivity for every c . In particular, any cyclic trace on a larger formal differential-operator algebra that restricts to this formula cannot be positive for an involution restricting to the one above. This conclusion needs no choice of an involution on the larger algebra.

The finite-dimensional corners $Q_n \mathcal{F}_0 Q_n$ do have positive traces, and the traces agree under their inclusions. Their units are Q_n , with masses n . Compatibility of these finite traces supplies τ on their union, but supplies no uniform mass bound on a unitization. The common GNS domain and a vector representing the functional are therefore separate constructions even in this explicit matrix example.

PROPOSITION 1.19.2 (*Formal completion and Hilbert adjoints*). Conjugate transpose on $\mathcal{F}_0$ does not extend to a continuous involution on $\mathcal{E}$ with its operator topology. The finite-matrix action on D_0 also does not extend by the same coefficient formula to an action of all $\mathcal{E}$ on $\ell^2(\mathbb{N})$.

Proof. Let $T_n = \sum_{i=0}^n E_{i0}$. Every fixed row stabilizes, so T_n converges in $\mathcal{E}$ to

$$T(h) = h_0(1, 1, 1, \dots).$$

The matrices $T_n^* = \sum_{i=0}^n E_{0i}$ do not converge in $\mathcal{E}$. Their zeroth rows do not stabilize, and an entire output row is discrete in the operator topology. Continuity of a proposed involution would send the convergent sequence T_n to a convergent sequence, which is impossible. For the second assertion, $e_0 \in \ell^2$ but $T(e_0) = (1, 1, \dots) \notin \ell^2$. $\square$

There is also a direct norm obstruction. The projections E_{nn} tend to zero in the formal operator topology, but each has Hilbert operator norm one and GNS vector norm one for τ . Thus the formal completion, operator-norm completion and GNS completion are different constructions on the same finite matrices. For the Weil algebra, Theorem 1.12.2 specifies a fourth topology when its multipliers are unbounded: the common graph topology. None of these completions is selected by cyclicity alone.

1.20 GAUSSIAN AVERAGES AND FORMAL COEFFICIENTS

The Gaussian in the explicit formula also permits an exact comparison between a numerical integral and formal polynomial elimination. It is necessary to specify both the observable algebra and the parameter. For a positive real number h , set

$$\mathbb{E}_h(p) = \frac{1}{\sqrt{2\pi h}} \int_{\mathbb{R}} p(u) e^{-u^2/(2h)} du = \int_{\mathbb{R}} p(u) g_{h/2}(u) du, \quad p \in \mathbb{C}[u].$$

The polynomial algebra here uses pointwise multiplication and $p^*(u) = \overline{p(u)}$ for real u . Its involution fixes u . Polynomial growth and Gaussian decay make every displayed integral absolutely convergent.

PROPOSITION 1.20.1 (*Numerical and formal Gaussian elimination*). For every polynomial p and real $h > 0$,

$$\mathbb{E}_h(p) = \sum_{m \geq 0} \frac{h^m}{2^m m!} p^{(2m)}(0). \quad (1.73)$$

The sum is finite. In particular, $\mathbb{E}_h$ is positive, $\mathbb{E}_h(u) = 0$ and $\mathbb{E}_h(u^2) = h$.

Let $\hbar$ instead be a formal central variable of degree zero. The same finite derivative formula on each polynomial gives a continuous $\mathbb{C}[[\hbar]]$ -linear map

$$G : \mathbb{C}[u][[\hbar]] \longrightarrow \mathbb{C}[[\hbar]], \quad G\left(\sum_{j \geq 0} p_j(u) \hbar^j\right) = \sum_{j, m \geq 0} \frac{p_j^{(2m)}(0)}{2^m m!} \hbar^{j+m}.$$

Continuity refers to the $\hbar$ -adic topologies. On $\mathbb{C}[u, \hbar]$, evaluating this result at $\hbar = h > 0$ agrees with the numerical Gaussian average of the specialized polynomial. There is no unital $\mathbb{C}$ -algebra homomorphism $\mathbb{C}[[\hbar]] \rightarrow \mathbb{C}$ that sends $\hbar$ to a nonzero number h .

Proof. Lemma 1.3.2 gives $\mathbb{E}_h(1) = 1$. Reflection makes every odd moment zero. For $n \geq 2$, integrate the derivative of $u^{n-1} e^{-u^2/(2h)}$ over $\mathbb{R}$. Its values at both ends are zero, and its derivative is integrable. It follows that

$$\mathbb{E}_h(u^n) = (n-1)h \mathbb{E}_h(u^{n-2}).$$

Thus $\mathbb{E}_h(u^{2m}) = (2m)! h^m / (2^m m!)$. Expansion of p proves (1.73). The positive integral of $|p|^2$ proves positivity, and the moments of u and u^2 have the stated values.

In the formal formula, the coefficient of $\hbar^N$ uses only $j, m \geq 0$ with $j + m = N$. There are finitely many such pairs. Each derivative is a derivative of a polynomial, so this defines every coefficient without a convergence assumption. The map preserves every power of $\hbar$, proving continuity. Reindexing these finite coefficient sums proves $\mathbb{C}[[\hbar]]$ -linearity. For a polynomial in $u, \hbar$, all sums are finite, so numerical specialization commutes with the proved moment formula.

For $h \neq 0$, the formal series

$$(1 - \hbar/h)^{-1} = \sum_{n \geq 0} (\hbar/h)^n$$

is an inverse in $\mathbb{C}[[\hbar]]$. A unital homomorphism sending $\hbar$ to h would send the unit $1 - \hbar/h$ to zero, which is impossible. At $h = 0$, the constant-coefficient homomorphism does exist. $\square$

The comparison at positive h therefore holds on the polynomial subalgebra. Extension to a selected class of infinite series requires an additional summability rule and its compatibility with the operations. Formal coefficientwise existence alone gives no such rule.

The Gaussian average is not multiplicative: $\mathbb{E}_h(u)^2 = 0$ while $\mathbb{E}_h(u^2) = h$. Nor does inserting a Gaussian turn pointwise polynomial multiplication into convolution. For $A > 0$, the linear map

$$J_A : \mathbb{C}[u] \longrightarrow \mathcal{A}_\zeta, \quad J_A(p)(u) = p(u)g_A(u)$$

is well-defined, since every derivative is a polynomial times a Gaussian. But

$$J_A(1) \star J_A(1) = g_{2A} \neq g_A = J_A(1), \quad J_A(u)^* = -J_A(u) \neq J_A(u^*).$$

The first inequality follows already from the values at zero. The second uses the reflection in the convolution involution. Thus J_A is neither an algebra homomorphism nor an involution-preserving map. Positivity of the numerical Gaussian average cannot pass through it by Theorem 1.14.1.

1.21 THE REMAINING COMPARISON WITH QUANTUM CURRENTS

The formal Hamiltonian bracket on H_{for} is $[f, g] = f_x g_y - f_y g_x$ modulo constants. Put $a_f = -f_y \partial_x + f_x \partial_y$ on this quotient. Changing a representative by a constant changes neither the derivatives nor the induced operator. An output through degree N uses only monomials through degree $N + 1$ in each input, because a bracket of monomials has degree equal to the sum of their degrees minus two. This proves continuity, including continuity of $f \mapsto a_f$ in the row topology. Computing the two derivative coefficients gives

$$[a_f, a_g] = -\partial_y[f, g] \partial_x + \partial_x[f, g] \partial_y = a_{[f, g]}.$$

Thus these operators give an actual symmetry action on H_{for} . A numerical positive representation needs further data: a complex Hilbert space, a common invariant dense domain, and operators whose restricted adjoints realize the chosen involution. Proposition 1.19.2 prevents obtaining these data by completing all finite matrix kernels in the formal operator topology. Proposition 1.20.1 also prevents specializing an arbitrary formal parameter at a positive numerical value. These are restrictions on the stated carriers and maps.

An algebra generated by products includes the empty product as a unit. For a positive comparison, one might choose an ordinary degree-zero subalgebra containing this identity and closed under an involution. The finite-mass calculation excludes this choice if the positive scalar functional is to be defined on the entire subalgebra.

PROPOSITION 1.21.1 (*An obstruction for unital observable algebras*). There is no unital complex $*$ -algebra $\mathfrak{B}$, positive linear functional $\omega : \mathfrak{B} \rightarrow \mathbb{C}$, and $*$ -homomorphism $T : \mathcal{A}_\zeta \rightarrow \mathfrak{B}$ such that $L_\zeta = \omega \circ T$. No continuity hypothesis on T or ω is needed.

Proof. Suppose these data exist. Put $m = \omega(1)$, a finite nonnegative real number. Extend T to the algebraic unitization by

$$T^+(f + \lambda 1) = T(f) + \lambda 1_{\mathfrak{B}}.$$

The unitization product and the multiplicativity of T show that T^+ is a unital $*$ -homomorphism. Thus $\omega \circ T^+$ is a positive extension of L_ζ with mass m . By (1.60),

$$|L_\zeta(g_a)|^2 \leq m L_\zeta(g_a^* \star g_a) = m L_\zeta(g_{2a}) \quad (a > 0).$$

Theorem 1.16.1 makes the ratio on the left to $L_\zeta(g_{2a})$ unbounded as a decreases to zero. The denominator is positive for all sufficiently small a , by the same asymptotic. This contradicts the finite bound m . $\square$

A remaining positive arithmetic comparison must therefore choose a proper nonunital algebra on which the functional is finite. Its first missing map is an involution-preserving algebra map

$$T : \mathcal{A}_\zeta \longrightarrow \mathfrak{B}$$

into such a specified algebra of current observables, together with a positive linear functional ω on $\mathfrak{B}$ satisfying $L_\zeta = \omega \circ T$. The algebra, involution, domain and positivity of ω must be constructed independently of the zeta zero locations. Theorem 1.14.1 would give the isometry $U : [f] \mapsto [T(f)]$ onto the closed subspace $K = \overline{[T(\mathcal{A}_\zeta)]} \subset \mathcal{H}_\omega$. Proposition 1.14.2 identifies the transported common graph domain with that of the restricted multipliers on $E = [T(\mathcal{A}_\zeta)] \subset K$, with every closure taken in K . Equality with the full ambient common domain requires a further hypothesis, such as surjectivity of T . The finite-mass obstruction in Theorem 1.16.1 still applies: a finite-norm vector representing this functional, or a positive functional on a C^* -algebra as target, cannot supply this factorization. A positive functional on a proper nonunital algebra remains a possible kind of carrier, as the finite-matrix trace demonstrates. This example does not supply the map T or the equality with L_ζ .

Any current-field construction must also specify its operator products, contraction tensors and the scalar trace or weight on those products. Their convergence with the spatial propagators is an additional analytic condition. Neither the strict zero strip nor the equality of the Gaussian moment formulas constructs this comparison or proves the positivity of the full Weil form.

1.22 POSITIVE ARITHMETIC FORMS FROM PRIME TRANSLATIONS

The negative prime coefficients admit a positive realization after an explicit diagonal compensation. The compensation is forced by positivity, even for a single prime. This gives a nonunital arithmetic carrier with infinite mass and determines the obstruction to summing its positive forms over all primes. The comparison uses the convolution algebra $\mathcal{A}_\zeta$ and the Fourier normalization of (1.35) throughout. All spaces and algebras in this section are complex and have degree zero. There is no formal parameter or change of coefficient field.

For $v \in \mathbb{R}$, let $(\tau_v f)(u) = f(u - v)$. Lebesgue measure makes τ_v unitary on $L^2(\mathbb{R}, du)$. If $f \in \mathcal{A}_\zeta$ and $g = f^* \star f$, direct substitution gives

$$g(o) = \|f\|_2^2, \quad g(v) = \langle f, \tau_v f \rangle_2, \quad \|f - \tau_v f\|_2^2 = 2g(o) - g(v) - g(-v). \quad (1.74)$$

The inner product is linear in its first argument. The reflected value satisfies $g(-v) = \overline{g(v)}$. These identities follow from absolutely convergent integrals, since $\mathcal{A}_\zeta \subset L^2(\mathbb{R})$.

Example 1.22.1 (One prime and one Gaussian). Retain only $n = 2$, with the native coefficient $c = \log 2/\sqrt{2}$ and translation length $\ell = \log 2$. Define

$$W_{\{2\}}(g) = c(2g(o) - g(\ell) - g(-\ell)).$$

Equation (1.74) gives $W_{\{2\}}(f^* \star f) = c\|f - \tau_\ell f\|_2^2 \geq 0$. For the normalized Gaussian $f = g_a$ with $a = \ell^2/8$,

$$\begin{aligned} W_{\{2\}}(g_{2a}) &= \sqrt{\frac{2}{\pi}}(1 - e^{-1}), \\ -c(g_{2a}(\ell) + g_{2a}(-\ell)) &= -\sqrt{\frac{2}{\pi}}e^{-1}, \\ 2c g_{2a}(o) &= \sqrt{\frac{2}{\pi}}. \end{aligned}$$

Indeed, $g_{2a}(o) = 1/(\sqrt{\pi}\ell)$ and $g_{2a}(\ell) = e^{-1}g_{2a}(o)$. Thus the raw prime term is already negative on this convolution square. Its positive realization includes the displayed nonzero diagonal term.

The same prime gives a two-dimensional calculation with an exact Gram matrix. Keep $a = \ell^2/8$, put $x = e^{-1}$, and use the ordered vectors $f_o = g_a, f_\ell = \tau_\ell g_a$. The Hermitian form $B(f, h) = W_{\{2\}}(h^* \star f)$ has matrix

$$G = \frac{1}{\sqrt{2\pi}} \begin{pmatrix} 2(1-x) & 2x-1-x^4 \\ 2x-1-x^4 & 2(1-x) \end{pmatrix}.$$

To compute its off-diagonal entry, expand $c\langle f_o - \tau_\ell f_o, \tau_\ell f_o - \tau_{2\ell} f_o \rangle_2$. The four correlations give $c(2g_{2a}(\ell) - g_{2a}(o) - g_{2a}(2\ell))$. The eigenvalues of $\sqrt{2\pi}G$ are

$$1 - x^4, \quad 3 - 4x + x^4 = (1-x)^2(x^2 + 2x + 3).$$

Both are positive because $0 < x < 1$. This calculation retains the actual prime weight, convolution involution, and off-diagonal pairing on the two displayed test functions.

Let $(c_n)_{n \geq 2}$ be nonnegative real numbers satisfying

$$0 < C := \sum_{n \geq 2} c_n < \infty.$$

Only indices with $c_n > 0$ contribute. The cases of interest retain the native weights $\Lambda_{\text{vM}}(n)/\sqrt{n}$ on finitely many prime powers, or on every power of finitely many primes. Define

$$\begin{aligned} W_c(g) &= \sum_{n \geq 2} c_n (2g(0) - g(\log n) - g(-\log n)), \\ \lambda_c(t) &= \sum_{n \geq 2} c_n |1 - e^{it \log n}|^2 = 2 \sum_{n \geq 2} c_n (1 - \cos(t \log n)), \\ d\mu_c(t) &= \lambda_c(t) \frac{dt}{2\pi}. \end{aligned} \tag{1.75}$$

The coefficient sum makes the first series absolutely convergent for bounded g , and the second uniformly convergent on $\mathbb{R}$. Consequently λ_c is continuous and $0 \leq \lambda_c \leq 4C$. Choosing any n_0 with $c_{n_0} > 0$ shows that its zeros belong to $\{2\pi k / \log n_0 : k \in \mathbb{Z}\}$. It is therefore positive almost everywhere. When all coefficients vanish, W_c and μ_c are zero. That degenerate case has a zero Hilbert space and admits finite positive unitizations. The hypothesis $C > 0$ excludes it in the following statements.

THEOREM 1.22.2 (*The arithmetic positive form and its least compensation*). For every $g \in \mathcal{A}_\zeta$ and $f \in \mathcal{A}_\zeta$,

$$\begin{aligned} W_c(g) &= \int_{\mathbb{R}} \widehat{g}(t) d\mu_c(t), \\ W_c(f^* \star f) &= \sum_{n \geq 2} c_n \|f - \tau_{\log n} f\|_2^2 = \int_{\mathbb{R}} |\widehat{f}(t)|^2 d\mu_c(t). \end{aligned} \tag{1.76}$$

The second value is strictly positive for $f \neq 0$. For real d , define

$$V_d(g) = d g(0) - \sum_{n \geq 2} c_n (g(\log n) + g(-\log n)).$$

Then $V_d(f^* \star f) \geq 0$ for every $f \in \mathcal{A}_\zeta$ if and only if $d \geq 2C$.

Proof. Fourier inversion gives

$$2g(0) - g(v) - g(-v) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) (2 - e^{-itv} - e^{itv}) dt.$$

Since $\widehat{g}$ is integrable and $\sum c_n = C$, an integrable bound $4C|\widehat{g}(t)|$ permits summation inside the integral. This proves the first identity in (1.76). Apply it to $g = f^* \star f$ and use $\widehat{f^* \star f} = |\widehat{f}|^2$ on the real axis. Equation (1.74) gives the other expression. If its value is zero, $\widehat{f}$ vanishes almost everywhere because $\lambda_c > 0$ almost everywhere. Continuity and Fourier inversion then give $f = 0$.

For $d \geq 2C$, the identity $V_d(f^* \star f) = W_c(f^* \star f) + (d - 2C)\|f\|_2^2$ proves positivity. Conversely, use $f = g_A$ and divide by the positive number $g_{2A}(0)$. The Gaussian formula gives

$$\frac{V_d(g_{2A})}{g_{2A}(0)} = d - 2 \sum_{n \geq 2} c_n e^{-(\log n)^2 / (8A)} \longrightarrow d - 2C \quad (A \rightarrow \infty).$$

The summable coefficients justify the limit. If $d < 2C$, the value is negative for sufficiently large A . $\square$

Every power of one prime can be retained without a cutoff. For a prime p , put $r = p^{-1/2}$, $\theta = t \log p$, and $c_{p^m} = \log p r^m$ for $m \geq 1$, with all other coefficients zero. Then

$$C_p = \frac{\log p r}{1 - r}, \quad \lambda_p(t) = \frac{2 \log p r (1 + r) (1 - \cos \theta)}{(1 - r)(1 - 2r \cos \theta + r^2)}. \tag{1.77}$$

To verify the expression, sum $\sum_{m \geq 1} r^m = r/(1-r)$ and take the real part of $\sum_{m \geq 1} (re^{i\theta})^m$. The latter real part is $(r \cos \theta - r^2)/(1 - 2r \cos \theta + r^2)$. Subtracting gives (1.77). Its denominator is positive because $0 < r < 1$. This constructs the positive form of one complete local Euler factor, with exactly its prime-power coefficients and least diagonal compensation.

1.22.1 A nonunital algebra and its Hilbert completion

Let $\mathcal{S}(\mathbb{R})$ be the space of smooth functions b such that

$$\sup_{t \in \mathbb{R}} (1 + |t|)^j |b^{(k)}(t)| < \infty \quad (j, k \geq 0).$$

These seminorms specify its topology. Use pointwise multiplication and the involution $b^*(t) = \overline{b(t)}$. The product rule makes $\mathcal{S}(\mathbb{R})$ a complex $*$ -algebra. It has no identity: a pointwise identity must equal one wherever e^{-t^2} is nonzero, whereas Schwartz functions tend to zero at infinity. Define

$$\mathfrak{B}_c = \mathcal{S}(\mathbb{R}), \quad \omega_c(b) = \int_{\mathbb{R}} b(t) d\mu_c(t), \quad T_c : \mathcal{A}_c \longrightarrow \mathfrak{B}_c, \quad T_c(g) = \widehat{g}|_{\mathbb{R}}.$$

The bound $\lambda_c \leq 4C$ makes this integral finite and continuous in the Schwartz topology. It is not asserted to be continuous in the supremum norm.

THEOREM 1.22.3 (Arithmetic GNS representation). The map T_c is an injective $*$ -homomorphism, and $W_c = \omega_c \circ T_c$. The functional ω_c is positive with zero GNS null space. Its GNS Hilbert space and algebraic invariant domain are

$$H_c = L^2(\mathbb{R}, \mu_c), \quad D_c = \mathcal{S}(\mathbb{R}) \subset H_c,$$

with action $(\pi_c(b)h)(t) = b(t)h(t)$. Every $\pi_c(b)$ extends boundedly, with $\|\pi_c(b)\| = \sup_{t \in \mathbb{R}} |b(t)|$. The image $T_c(\mathcal{A}_c)$ is dense in H_c . Thus the GNS isometry $[f] \mapsto \widehat{f}$ for W_c is onto after Hilbert completion. Neither W_c nor ω_c admits a positive unitization of finite mass.

Proof. For $g \in \mathcal{A}_c$, every derivative of $u^j g(u)$ is integrable. Fourier differentiation and repeated integration by parts therefore show $\widehat{g} \in \mathcal{S}(\mathbb{R})$. The product and involution identities in (1.34) prove that T_c is a $*$ -homomorphism. Fourier inversion proves its injectivity, and (1.76) proves the equality of functionals. The identity

$$\omega_c(b^*b) = \int_{\mathbb{R}} |b(t)|^2 d\mu_c(t)$$

proves positivity and identifies its null space as zero.

Smooth compactly supported functions are dense in H_c . Indeed, truncate a vector's support and size first. The resulting bounded functions of compact support belong to ordinary $L^2(dt)$ and approximate the vector in H_c by dominated convergence. Approximate them in $L^2(dt)$ by smooth compactly supported functions. The bound $d\mu_c \leq (2C/\pi)dt$ transfers this approximation to H_c . For this ordinary approximation, step functions are dense in $L^2(dt)$, and smooth ramps replace their interval endpoints on sets of vanishing measure. It follows that D_c has Hilbert completion H_c . Multiplication preserves D_c and satisfies the adjoint identity there. Its norm is at most $\sup |b|$. Every interval has positive μ_c measure. A bump function in an interval where $|b| > \sup |b| - \epsilon$ gives the reverse norm bound, after normalization and then $\epsilon \downarrow 0$.

To prove density of the image, take $b \in C_c^\infty(\mathbb{R})$ and let f be its inverse Fourier transform. Integration by parts shows that $f \in \mathcal{S}(\mathbb{R})$ and $\widehat{f} = b$. Choose a smooth cutoff χ equal to one on $[-1, 1]$ and supported in $[-2, 2]$. Then $f_j(u) = \chi(u/j)f(u)$ belongs to $C_c^\infty(\mathbb{R}) \subset \mathcal{A}_\zeta$ and converges to f in every Schwartz seminorm. Fourier differentiation and integration by parts show that $\widehat{f_j} \rightarrow b$ in every Schwartz seminorm. In particular, this convergence holds in H_c . The preceding density proves the assertion about the GNS isometry.

Finally, evaluate W_c on the Gaussians g_a as $a \downarrow 0$. Since every contributing n is at least two,

$$\begin{aligned} W_c(g_a) &= \frac{1}{\sqrt{\pi a}} \sum_{n \geq 2} c_n (1 - e^{-(\log n)^2/(4a)}) \\ &= \frac{C}{\sqrt{\pi a}} \left(1 + O\left(e^{-(\log 2)^2/(4a)}\right) \right), \\ \frac{|W_c(g_a)|^2}{W_c(g_{2a})} &\sim \frac{\sqrt{2} C}{\sqrt{\pi}} a^{-1/2} \longrightarrow \infty. \end{aligned} \tag{1.78}$$

The unitization inequality (1.60) excludes any finite mass. A positive unitization of ω_c would pull back through T_c to one of W_c , so it is excluded as well. $\square$

The weight can also be defined on the positive cone of $L^\infty(\mu_c)$ by $\Phi_c(b) = \int b d\mu_c \in [0, \infty]$. Its finite linear ideal is $\mathfrak{m}_c = L^1(\mu_c) \cap L^\infty(\mu_c)$. This ideal is closed under conjugation and multiplication by bounded functions. Its square-integrable ideal is $\mathfrak{n}_c = L^2(\mu_c) \cap L^\infty(\mu_c)$. For $b \in \mathfrak{n}_c$, one has $b^*b \in \mathfrak{m}_c$ and $\Phi_c(b^*b) = \|b\|_{H_c}^2$. Bounded functions supported on finite intervals approximate any vector in H_c . Moreover, restricting a nonnegative bounded function to these intervals gives increasing finite-weight functions with that function as their limit. These statements specify the finite ideals without assigning a finite value to the identity. Equation (1.78) implies $\Phi_c(1) = \infty$.

Although every Schwartz multiplier is bounded, the functional has no finite-norm representing vector in any invariant $*$ -representation domain. Such a vector would give $|W_c(g)|^2 \leq \|\Omega\|^2 W_c(g^* \star g)$ by Cauchy–Schwarz, contrary to (1.78). The positive functional is defined on the proper algebra $\mathfrak{B}_c$. It does not extend positively to its entire C^* completion. Such an extension would admit a finite positive unitization by Theorem 1.15.1, contrary to the proved divergence.

1.22.2 Frequency multipliers on a common invariant domain

There is a natural unbounded multiplier compatible with this arithmetic representation: the real Fourier coordinate t . For clarity, it is a multiplier of $\mathfrak{B}_c$, not an element of it. Its powers cannot be inserted as finite-weight observables. Put

$$\begin{aligned} (Nh)(t) &= th(t), & \mathcal{D}(N) &= \{h \in H_c : th \in H_c\}, \\ D_c^\infty &= \bigcap_{k \geq 0} \{h \in H_c : t^k h \in H_c\}. \end{aligned} \tag{1.79}$$

PROPOSITION 1.22.4 (*Self-adjoint frequency and all products*). The operator N is unbounded and self-adjoint. Schwartz functions form a core for every power N^k . The space D_c^∞ is dense and complete for the seminorms $\|t^k h\|_{H_c}$, $k \geq 0$. It is invariant under N , polynomial multipliers, Schwartz multipliers, and $U_s h(t) = e^{ist} h(t)$ for every real s . All products and adjoint identities hold on this common domain. The image $T_c(\mathcal{A}_\zeta)$ is dense in D_c^∞ for these seminorms. On that image, $NT_c(f) = T_c(i\partial_u f)$ and $U_s T_c(f) = T_c(\tau_s f)$.

Proof. Multiplication by a measurable real function on its maximal domain is closed in this case: take an almost-everywhere convergent subsequence of a convergent pair $h_j \rightarrow h$, $th_j \rightarrow v$ in H_c . Its limits obey $v = th$. To construct such a subsequence, choose successive errors with summable squared norms and apply the integral bound to their squared sum.

The operator is symmetric. If v belongs to its adjoint domain, the adjoint identity tested on all bounded functions of compact support forces $N^*v = tv$ almost everywhere. Those tests belong to $\mathcal{D}(N)$, and on each compact interval both sides are integrable against them by Cauchy–Schwarz. Thus $tv \in H_c$ and $v \in \mathcal{D}(N)$. This proves self-adjointness. The same argument applies to multiplication by t^k . Support and size truncation followed by smooth approximation gives density of C_c^∞ in its graph norm, because $(1 + t^{2k})\lambda_c(t)$ is bounded on each compact interval. It follows that Schwartz functions form a core. Normalized bumps supported in $(R, R + 1)$ have $\|Nh\| \geq R\|h\|$, so N is unbounded.

A Cauchy sequence in all the displayed seminorms has a Hilbert limit h and limits for every $t^k h_j$. Closedness of each multiplication operator identifies these limits as $t^k h$ and proves completeness. Support truncation and smooth approximation give density of Schwartz functions simultaneously in any finite collection of the seminorms. Polynomial multiplication only increases their indices. A Schwartz multiplier b obeys $\|t^k b h\| \leq \|b\|_\infty \|t^k h\|$, and U_s preserves each seminorm. These estimates prove invariance and continuity. Pointwise multiplication and conjugation prove all product and adjoint identities on D_c^∞ .

For $b \in \mathcal{S}$, the cutoff construction in the preceding theorem approximates b by transforms of compactly supported smooth functions in the Schwartz topology. That topology controls every seminorm in (1.79), since λ_c is bounded. This proves the image density. Integration by parts gives $i\widehat{\partial_u f}(t) = t\widehat{f}(t)$, and a change of variable gives $\widehat{\tau_s f}(t) = e^{ist}\widehat{f}(t)$. Both $i\partial_u f$ and $\tau_s f$ remain in $\mathcal{A}_\zeta$. $\square$

Dominated convergence makes $(U_s)_{s \in \mathbb{R}}$ a strongly continuous unitary group on H_c . For $h \in \mathcal{D}(N)$, the estimate $|(e^{ist} - 1)/s| \leq |t|$ gives $(U_s h - h)/s \rightarrow iN h$ in H_c . Conversely, a convergent difference quotient has almost-everywhere limit ith along a subsequence, so its Hilbert limit requires $th \in H_c$. Thus its generator domain is exactly $\mathcal{D}(N)$. The smaller domain D_c^∞ permits every polynomial in this generator. For the Schwartz algebra alone, its common GNS graph completion is all of H_c , since each of its multipliers is bounded. These two domain statements refer to different multiplier algebras.

1.22.3 The sum over all primes and the native comparison

For $X \geq 2$ and a real exponent σ , set

$$C_{X,\sigma} = \sum_{2 \leq n \leq X} \Lambda_{\text{vM}}(n) n^{-\sigma}, \quad q_{X,\sigma}(f) = \sum_{2 \leq n \leq X} \Lambda_{\text{vM}}(n) n^{-\sigma} \|f - \tau_{\log n} f\|_2^2.$$

These forms are defined for every $f \in L^2(\mathbb{R}, du)$. Increasing X increases their values. The native exponent is $\sigma = 1/2$.

THEOREM 1.22.5 (*The exact convergence boundary for the positive prime sum*). For $\sigma > 1$, the coefficient sum $C_\sigma = \sum_{n \geq 2} \Lambda_{\text{vM}}(n)n^{-\sigma}$ is finite. It therefore defines the positive carrier above with all primes present. Moreover,

$$0 \leq q_\sigma(f) := \lim_{X \rightarrow \infty} q_{X,\sigma}(f) \leq 4C_\sigma \|f\|_2^2.$$

For every real $\sigma \leq 1$ and every nonzero $f \in L^2(\mathbb{R})$,

$$\lim_{X \rightarrow \infty} q_{X,\sigma}(f) = +\infty. \quad (1.80)$$

Thus the finite domain of the increasing limit form is $\{0\}$ when $\sigma \leq 1$. It is not a densely defined quadratic form on $L^2(\mathbb{R})$.

Proof. For $\sigma > 1$, use $0 \leq \Lambda_{\text{vM}}(n) \leq \log n$ and the integral comparison for $\sum (\log n)n^{-\sigma}$. The bound $\|f - \tau_v f\|_2 \leq 2\|f\|_2$ proves the stated estimate.

For divergence of the coefficients, prime factorization gives $\sum_{d|m} \Lambda_{\text{vM}}(d) = \log m$. For an integer $M \geq 4$, sum this identity over $m \leq M$ to obtain

$$\log(M!) = \sum_{n=2}^M \Lambda_{\text{vM}}(n) \left\lfloor \frac{M}{n} \right\rfloor \leq M \sum_{n=2}^M \frac{\Lambda_{\text{vM}}(n)}{n}.$$

The last $\lceil M/2 \rceil$ factorial factors are at least $M/2$. Hence the last sum is at least $\frac{1}{2} \log(M/2)$ and tends to infinity. For $\sigma \leq 1$, $n^{-\sigma} \geq n^{-1}$, so $C_{X,\sigma} \rightarrow \infty$ as well.

Finally, $\langle f, \tau_v f \rangle_2 \rightarrow 0$ as $v \rightarrow +\infty$. To prove this, approximate f in L^2 by a function h supported in a fixed finite interval. For sufficiently large v , h and $\tau_v h$ have disjoint supports. Cauchy–Schwarz bounds the difference between their correlation and that of f by $\|f - h\|_2(\|f\|_2 + \|h\|_2)$, uniformly in v . The approximation error can tend to zero. It follows that $\|f - \tau_{\log n} f\|_2^2 \rightarrow 2\|f\|_2^2$ as $n \rightarrow \infty$. For $f \neq 0$, these squared norms are at least $\|f\|_2^2$ for every sufficiently large n . The divergent coefficient tail proves (1.80). $\square$

The damping exponent changes the coefficients and therefore changes the functional. For $\sigma > 1$, the Euler product in (1.9) also writes its positive Fourier density as

$$\lambda_\sigma(t) = 2 \left(-\frac{\zeta'}{\zeta}(\sigma) + \operatorname{Re} \frac{\zeta'}{\zeta}(\sigma - it) \right).$$

This equality uses only absolutely convergent Dirichlet series. Continuation of its right side beyond that half-plane does not continue the positive sum in (1.75). Theorem 1.22.5 proves the failure of that sum already at $\sigma = 1$.

For the exact comparison with L_ζ , put $C_X = C_{X,1/2}$ and let W_X have coefficients $c_n = \Lambda_{\text{vM}}(n)/\sqrt{n}$ for $n \leq X$, zero otherwise. Retain the polar and archimedean functional

$$R_\infty(g) = \widehat{g}(i/2) + \widehat{g}(-i/2) + \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) [\operatorname{Re} \psi(1/4 + it/2) - \log \pi] dt.$$

PROPOSITION 1.22.6 (*Subtraction and the remaining arithmetic form*). For every $g \in \mathcal{A}_\zeta$,

$$\begin{aligned} L_\zeta(g) &= W_X(g) + R_\infty(g) - 2C_X g(o) \\ &\quad - \sum_{n>X} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (g(\log n) + g(-\log n)). \end{aligned} \quad (1.81)$$

The tail tends to zero as $X \rightarrow \infty$. Consequently

$$\lim_{X \rightarrow \infty} (W_X(g) - 2C_X g(o)) = - \sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (g(\log n) + g(-\log n)).$$

The functional on the right is strictly negative on every Gaussian convolution square. For each nonzero $f \in \mathcal{A}_\zeta$,

$$R_\infty(f^* \star f) - 2C_X \|f\|_2^2 \longrightarrow -\infty. \quad (1.82)$$

Thus the full functional cannot be obtained as the increasing limit of these positive prime forms, or by adding a positive residual to each sufficiently large cutoff.

Proof. Substitute the definition of W_X into (1.39). This gives (1.81) with its stated signs. Choose $a > 1/2$ such that $g \in \mathcal{A}_a$. Lemma 1.3.1 bounds the absolute tail by

$$K_g \sum_{n>X} (\log n) n^{-a-1/2},$$

which tends to zero because $a + 1/2 > 1$. The Gaussian g_{2b} is strictly positive at every real argument. At least the $n = 2$ term is nonzero, so the negative prime sum on $g_{2b} = g_b^* \star g_b$ is strictly negative. The value $R_\infty(f^* \star f)$ is finite and independent of X . Since $C_X \rightarrow \infty$ and $\|f\|_2 > 0$, (1.82) follows. The full residual $L_\zeta - W_X$ differs from this expression on the chosen square by a tail tending to zero. It is therefore negative there for every sufficiently large X . $\square$

The construction gives an arithmetic $*$ -homomorphism and a positive functional with the required failure of finite mass. Its pullback is W_c , with the exact diagonal compensation specified above. For every fixed summable coefficient family, the Gaussian asymptotics also distinguish this pullback from the full Weil functional:

$$\frac{W_c(g_a)}{L_\zeta(g_a)} \sim \frac{4C}{\log(1/a)} \longrightarrow 0 \quad (a \downarrow 0).$$

This follows by dividing (1.78) by (1.66). The equality $L_\zeta = \omega \circ T$ still requires a construction that combines the prime, polar, and archimedean terms before asserting positivity. Equation (1.81) states the entire missing comparison. A representation of quantum currents would additionally have to identify their products and involution with this numerical carrier on a common domain. No formal completion, change of exponent, or removal of a divergent diagonal term supplies either identification.